12.06 .2026 Wrocław

PROGRAMOWANIE OBIEKTOWE

Projekt

Wydział Informatyki i Telekomunukacji	**Kierunek: Informatyka Techniczna**
Grupa zajęciowa (np. Pn7:30):Czw 13:15	**Semestr: 2025/26 LATO**
Nazwisko i Imię:Filip Klich	**Nr indeksu:293446**
Nazwisko i Imię:Michał Sikora	**Nr indeksu:292845**
Nazwisko i Imię:	**Nr indeksu:**
Nr. grupy projektowej:3	
Prowadzący: mgr inż. Karol Puchała	

TEMAT:

„Objectively Worse Doom”

OCENA: PUNKTY:

Data:

1. Spis narzędzi użytych przy tworzeniu projektu

- GCC 16.1.1 2026.04.30, w tym `g++`
- POSIXowy shell (`bash`)
- Coreutils, a dokladniej:
  - `find`
  - `echo`
  - `mkdir`
  - `rm`
  - `cat`
  - `[` (do sprawdzania kompatybilnosci)
  - `stat` (do sprawdzania kompatybilnosci)
  - `basename` (do sprawdzania kompatybilnosci)
  - `head` (do sprawdzania kompatybilnosci)
  - `mktemp` (do sprawdzania kompatybilnosci)
- GNU `sed`
- `lsof` (do sprawdzania kompatybilnosci)
- `objcopy`
- GNU `make`
- `expect` (do testow)
- libasound (ALSA)
- glibc

Dodatkowo:

- format object fileow to `elf64-x84-64`
- kernel z DRM/KMS oraz evdev

2. Założenia i opis funkcjonalny programu

2.1 Opis założeń

Projekt ma na celu stworzenie w pełni funkcjonalnej, minimalistycznej gry typu retro FPS. Wszystkie kluczowe elementy mechaniki gry (potwory, bronie) są projektowane z wykorzystaniem polimorfizmu, dziedziczenia i kompozycji, tak aby kod był czytelny i rozszerzalny.

- Gra jest w pełni obiektowa - każdy byt w świecie gry (potwór, broń, pocisk, gracz) jest instancją odpowiedniej klasy lub interfejsu.
- Polimorfizm jest używany jako główny mechanizm rozróżniania zachowania (różne typy potworów i pocisków).
- Gra działa w czasie rzeczywistym (pętla gry z update'ami i renderowaniem).
- Celem gry jest zabicie bossa.
- Prosta grafika imitujaca 3d, podobna do Castle Wolfenstein 3d, ale napisana od zera

---

2.2 Opis funkcjonalny

2.2.1 Ogólna koncepcja rozgrywki

Gracz wciela się w przeniesionego do innego swiata mieszkanca Japonii. Jego celem jest pokonanie armii krola demonow atakujacego krolestwo Aliud Regnum Mundi. Zadaniem jest przetrwanie i eliminacja wszystkich demonow oraz ich krola. Gracz może poruszać się, strzelać i zmieniać broń.

2.2.2 System potworów

Wszystkie potwory dziedziczą po wspólnej klasie abstrakcyjnej. Dzięki polimorfizmowi każdy typ zachowuje się inaczej, ale jest traktowany przez silnik gry w identyczny sposób.

2.2.2.1 Dostępne typy potworów

- **Podstawowy słaby** - najprostszy przeciwnik, mała ilość HP, atak w zwarciu
- **Slaby ranged** - tutorialowy przeciwnik strzelajacy
- **Assault shooter** - strzela seriami, średni zasięg
- **Sniper** - strzela rzadko, ale bardzo celnie i z dużej odległości
- **Duży gruby** - duża ilość HP, wolny, zadaje duże obrażenia w zwarciu
- **Mały szybki** - mały, bardzo szybki, bardzo mała ilość HP
- **All-rounder** - przeciwnik dobry zarowno w CQC, jak i w dystansie
- **Magic** - utility ale dla przeciwników
- **Elite Tank** - bardzo trudny przeciwnik do zabicia
- **Elite Szybki** - bardzo szybki przeciwnik ktory szarzuje
- **Boss**

2.2.3 System broni

Gracz może nosić wiele broni, zaimplementowanych polimorfizmem.

2.2.3.1 Dostępne bronie

- **M1911** - pistolet podstawowy, szybki słaby ogień pojedyńczy
- **Pistolet maszynowy** - szybkostrzelny, niskie obrażenia od trafienia
- **Karabin** - zrównoważona broń średniego dystansu
- **Sniper rifle** - bardzo wysokie obrażenia, wolne przeładowanie
- **Plasma gun** - strzela dużymi, powolnymi pociskami energetycznymi
- **Shotgun** - strzela wiązką śrutu (kilka pocisków naraz)
- **Katana** - broń biała, nie zużywa amunicji

2.2.4 System pocisków

Wszystkie pociski dziedziczą po klasie abstrakcyjnej `Projectile`

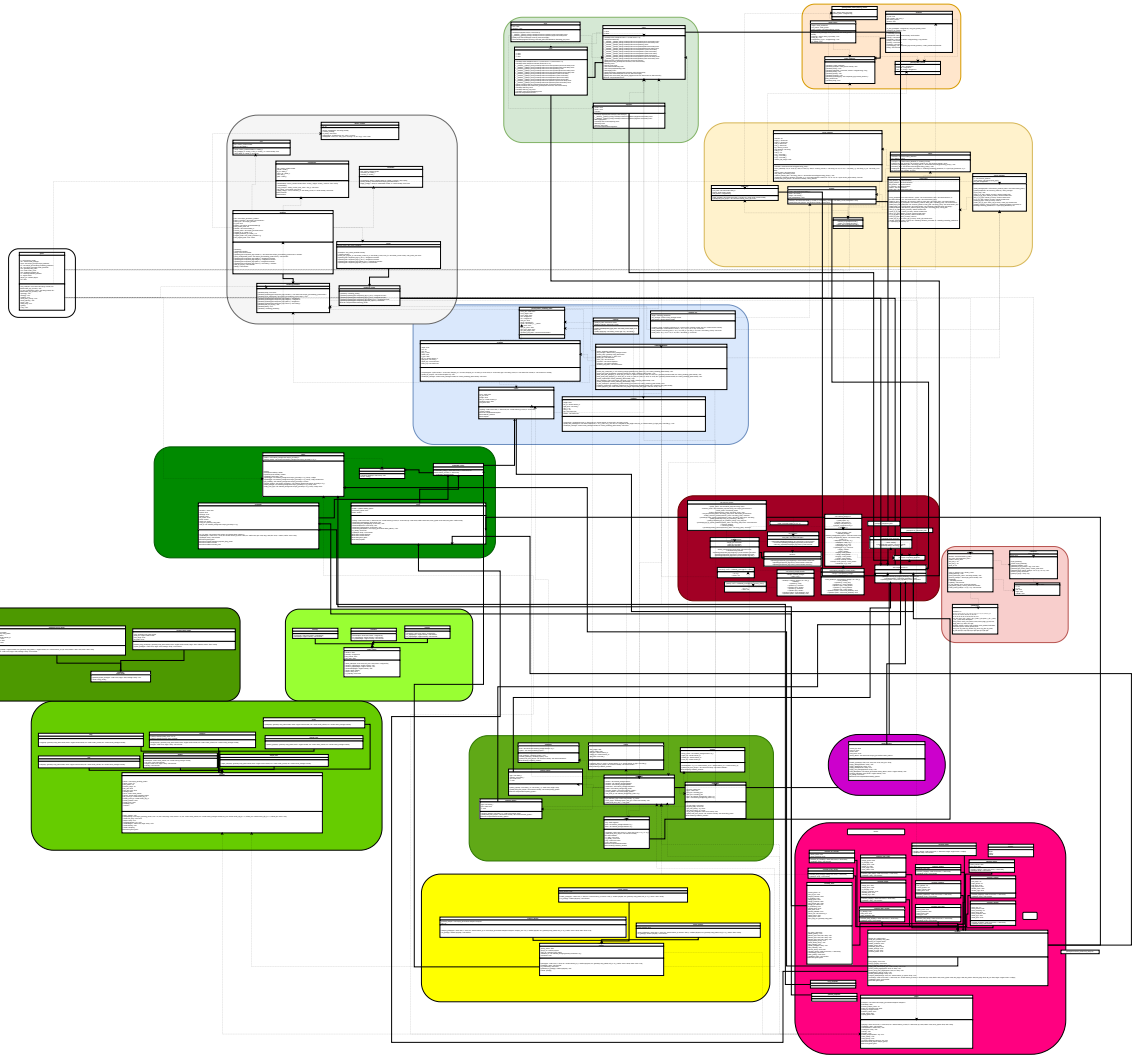
2.2.4.1 Typy pocisków

- **Pocisk zwyczajny** - klasyczny raycast
- **Plasma** - nie raycast,duże obrażenia, pocisk porusza się wolno
- **Slug** - wiele małych pocisków jednocześnie, nie raycast
- **Granat** - nie raycast, obrażenia obszarowe

2.2.5 Dodatkowe mechaniki

- System zdrowia i pancerza
- Proste pickupy
- Status effecty
- Efekty podpalenia i spowolnienia

3. Klasy



input/evdev/backend.h

```
#pragma once
#include <vector>
#include <bitset>
#include <linux/input.h>
#include "input/input-backend.h"

namespace input {
    namespace evdev {
        class backend : public input::input_backend {
        private:
            /* a single open evdev file */
            struct device_node {
                int fd;
                bool is_keyboard;
                bool is_mouse;
            };

            /* all open files */
            std::vector<device_node> devices;
            /* currently pressed keys */
            std::bitset<KEY_CNT> key_map;
            /* current mouse state */
            mouse_state mouse;

            /* last absolute state for EV_ABS */
            int last_abs_x;
            int last_abs_y;

            /* epoll file descriptor for reading only changed files */
            int epoll_fd;

            /* emulated-to-native map */
            uint16_t map_to_native(key k) const;
            /* probe all evdev devices */
            void probe_devices();
            /* probe a single evdev device */
            void probe_device(std::string const& dev_name);
            /* process a single struct input_event from evdev files */
            void process_event(struct input_event const& ev);

        public:
            /* constructor */
            backend();
            /* destructor */
            ~backend() override;

            /* update internal state */
            void update() override;
            /* read key from internal state */
            bool is_key_down(key k) const override;
            /* read internal mouse state */
            mouse_state get_mouse_state() const override;
            /* reset internal mouse state */
            void reset_mouse_state(int nx, int ny) override;
        };
```

```
    }
}
```

input/input-backend.h

```
#pragma once

#include <cstdint>

namespace input {
    /* recognized key map */
    enum class key {
        unknown = 0,
        a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z,
        n0, n1, n2, n3, n4, n5, n6, n7, n8, n9,
        f1, f2, f3, f4, f5, f6, f7, f8, f9, f10, f11, f12,
        esc, tab, caps_lock, l_shift, l_ctrl, l_meta, l_alt, space, r_alt, r_meta,
        r_ctrl, r_shift, enter, backspace,
        print_screen, scroll_lock, pause, insert, home, page_up, del, end, page_down,
        up, down, left, right,
        backtick, hyphen, equals, open_bracket, close_bracket, backslash, semicolon,
        quote, comma, period, slash,
        num_lock, kp_slash, kp_asterisk, kp_minus, kp_plus, kp_enter, kp_dot, kp0, kp1,
        kp2, kp3, kp4, kp5, kp6, kp7, kp8, kp9
    };

    /* recognized mouse state */
    struct mouse_state {
        int x, y;
        bool left, right, middle;
    };

    class input_backend {
    protected:
        /* bad flag, as in eg. std::filesystem::* */
        bool bad;
    public:
        /* constructor */
        input_backend();
        /* virtual destructor */
        virtual ~input_backend() = default;

        /* pure virtual "tick" method */
        virtual void update() = 0;
        /* pure virtual read from state */
        virtual bool is_key_down(key k) const = 0;
        /* pure virtual read from state */
        virtual mouse_state get_mouse_state() const = 0;
        /* pure virtual reset state */
        virtual void reset_mouse_state(int nx = 0, int ny = 0) = 0;
        /* virtual getter (we forgot about this one), as in std::filesystem */
        virtual bool is_bad() const;

        /* char-to-keymap helper */
        static key to_key(char c);
    };
}
```

rendering/drm-kms/crtc.h

```
#pragma once

#include <cstdint>
#include "device-context.h"
#include "mode.h"

namespace rendering {
    namespace drm_kms {
        class crtc {
            /* device owner */
            device_context const& dev;
            /* kernel CRTC id */
            std::uint32_t crtc_id;

        public:
            /* constructor */
            crtc(device_context const& d, uint32_t id);

            /* DRM_IOCTL_MODE_SETCRTC wrapper */
            bool set_config(uint32_t fb_id, uint32_t conn_id, mode const& m);

            /* DRM_IOCTL_MODE_PAGE_FLIP wrapper */
            bool page_flip(uint32_t fb_id) const;
        };
    }
}
```

rendering/drm-kms/framebuffer.h

```
#pragma once

#include <cstdint>
#include <cstddef>
#include "device-context.h"
#include "mode.h"

namespace rendering {
    namespace drm_kms {
        class crtc;
        class connector;
        class backend;

        class framebuffer {
            /* device owner */
            device_context const& dev;
            /* kernel framebuffer handle */
            uint32_t handle;

            /* kernel framebuffer id */
            uint32_t fb_id;
            /* mmap-ped memory pointer */
            uint32_t* mmio_ptr;
            /* mmap-ped memory size */
            size_t size;
            /* framebuffer pitch */
            uint32_t pitch;

        public:
            /* constructor */
            framebuffer(device_context const& d, uint32_t width, uint32_t height,
                bool *const success = nullptr);
            /* destructor */
            ~framebuffer();

            /* blit */
            void copy_from(uint32_t const* src, size_t pixel_count) const;

            /* page flip */
            bool flip_onto(crtc const& c) const;

            /* apply contfig to CRTC */
            bool apply_config(crtc &c, std::uint32_t const connector_id,
                mode const& m) const;

            friend connector;
            friend backend;
        };
    }
}
```

rendering/drm-kms/mode.h

```
#pragma once

#include <drm/drm.h>
#include "rendering/rendering-mode.h"

namespace rendering {
    namespace drm_kms {
        class mode : public rendering::rendering_mode {
            /* kernel drm_mode_modeinfo struct */
            struct drm_mode_modeinfo kernel_mode;

        public:

            /* constructor */
            mode(struct drm_mode_modeinfo const& info);
            /* destructor override with default */
            ~mode() override = default;

            /* CRTC creation request for ioctl */
            struct drm_mode_crtc create_crtc_req(std::uint32_t const crtc_id,
                std::uint32_t const fb_id, std::uint32_t const *const conn_id) const;

            /* componentized-like interface override */
            unsigned int operator()(util::component_tag<"x_res">) const override;
            /* componentized-like interface override */
            unsigned int operator()(util::component_tag<"y_res">) const override;
            /* componentized-like interface override */
            unsigned int operator()(util::component_tag<"refresh_hz">) const override;
            /* componentized-like interface override */
            bool operator()(util::component_tag<"has_vsync">) const override;
        };
    }
}
```

rendering/drm-kms/connector.h

```
#pragma once

#include <vector>
#include <memory>
#include <cstdint>
#include "device-context.h"
#include "mode.h"

namespace rendering {
    namespace drm_kms {
        class crtc;
        class framebuffer;

        class connector {
            /* device owner */
            device_context const& dev;
            /* kernel connector id */
            uint32_t connector_id;
            /* kernel encoder id */
            uint32_t encoder_id;

        public:
            /* constructor */
            connector(device_context const& d, uint32_t id, bool* const success = nullptr);

            /* returns std::vector of available modes for a particular connector */
            std::vector<std::unique_ptr<mode const>> probe_modes();

            /* config for ioctl */
            bool apply_config(crtc& c, framebuffer const& fb, mode const& m) const;
        };
    }
}
```

rendering/drm-kms/device-context.h

```
#pragma once

#include <string>

namespace rendering {
    namespace drm_kms {
        class device_context {
            /* dri file descriptor */
            int fd;

        public:

            /* constructor */
            explicit device_context(std::string const& path);
            /* destructor */
            ~device_context();

            /* error checking */
            bool is_valid() const;
            /* sys/ioctl.h ::ioctl wrapper */
            int ioctl(unsigned long request, void* arg) const;
            /* sys/mman.h ::mmap wrapper */
            void *mmap(void *addr, size_t len, int prot, int flags, off_t off) const;
        };
    }
}
```

rendering/drm-kms/backend.h

```
#pragma once

#include <memory>
#include <vector>
#include <drm/drm.h>

#include "rendering/rendering-backend.h"
#include "device-context.h"
#include "framebuffer.h"
#include "connector.h"
#include "crtc.h"

namespace rendering {
    namespace drm_kms {
        class backend : public rendering::rendering_backend {
        protected:
            /* opened /dev/dri device */
            std::unique_ptr<device_context> dev;
            /* currently active connector */
            std::unique_ptr<connector> active_connector;
            /* currently active crtc */
            std::unique_ptr<crtc> active_crtc;
            /* error checking */
            bool is_bad;

            /* double-buffering */
            std::unique_ptr<framebuffer> buffers[2];
            /* currently active buffers index */
            int front_buffer_index;
            /* shadow buffer */
            std::vector<uint32_t> shadow;

            /* currently chosen mode */
            std::unique_ptr<mode const> current_mode;

            /* framebuffer id for tty VT driver */
            uint32_t original_fb_id = 0;
            /* connector id for tty VT driver */
            uint32_t original_connector_id = 0;
            /* mode for tty VT driver */
            struct drm_mode_modeinfo original_mode = {};
            /* whether or not original VT state was preserved */
            bool has_original_state = false;

        public:
            /* constructor */
            backend();
            /* destructor override */
            ~backend() override;

            /* backend + fd aggregate */
            bool bad() const override;
            /* componentized-like interface override */
            virtual std::vector<std::unique_ptr<rendering_mode const>>
                operator()(util::component_tag<"modes">) override;
```

```
            /* virtual method override */
            void push_mode(std::unique_ptr<rendering_mode const> pushed_mode) override;
            /* componentized-like interface override */
            unsigned int operator()(util::component_tag<"width">) override;
            /* componentized-like interface override */
            unsigned int operator()(util::component_tag<"height">) override;
            /* componentized-like interface override */
            unsigned int operator()(util::component_tag<"pitch">) override;
            /* componentized-like interface override */
            std::uint32_t *operator()(util::component_tag<"mmio">) override;
            /* virtual method override */
            void wait_for_vsync() override;
            /* virtual method override */
            void flush() override;
        };
    }
}
```

rendering/renderer-2d.h

```
#pragma once


#include "rendering-backend.h"
#include "assets/asset-manager.h"
#include "assets/texture.h"
#include <string_view>
#include <cstdint>
#include <algorithm>

namespace rendering {
    class renderer_2d {
        /* chosen backend */
        rendering_backend &target;
        /* asset mgr ref */
        assets::asset_manager const& tex_manager;
        /* chosen 16x16 font atlas */
        assets::texture const& font_texture;

    public:
        /* constructor */
        renderer_2d(rendering_backend &tgt, assets::asset_manager const& tm,
            assets::texture const& font_tex);

        /* blit texture onto backend */
        void draw_texture(assets::texture const& tex, int x, int y, int w, int h) const;

        /* draw parts of font atlas onto backend, forming a string */
        void draw_text(std::string_view text, int x, int y, int char_w,
            int char_h, std::uint32_t const color) const;

        /* draw rectangle with alpha */
        void draw_rect(int x, int y, int w, int h, std::uint32_t color) const;
    };
}
```

rendering/lighting.h

```
#pragma once

#include <cstdint>

namespace rendering {
    class lighting {
        /* dynamic lighting minimum distance */
        static constexpr float dl_start   = 64.0f;
        /* dynamic lighting units per length */
        static constexpr float dl_density = 0.25f;

    public:

        /* calculate light level from sector light and z */
        __attribute__((always_inline)) static inline int
            calculate(std::uint8_t const sector_light_level, float const depth) {
            if(depth > dl_start) {
                int corrected_light = sector_light_level - (depth - dl_start) * dl_density;
                if(corrected_light < 0) corrected_light = 0;
                return corrected_light;
            } else {
                return sector_light_level;
            }
        }

        /* apply lighting to color using SWAR */
        __attribute__((always_inline)) static inline std::uint32_t
            apply(std::uint32_t const orig, int light) {
            std::uint32_t swar_rb = orig & 0x00ff00ff;
            std::uint32_t swar_g  = orig & 0x0000ff00;

            swar_rb = ((swar_rb * light) >> 8) & 0x00ff00ff;
            swar_g  = ((swar_g  * light) >> 8) & 0x0000ff00;

            return swar_rb | swar_g;
        }
    };
}
```

rendering/frd.h

```
#pragma once

#include <cstdint>
#include <vector>
#include "math/vec2.h"

namespace rendering {
    struct frame_rendering_data {
        /* camera position */
        math::vec2 cam_pos;
        /* camera height and angle */
        float cam_height, cam_angle;
        /* screen width and height */
        unsigned int sw, sh;
        /* half of screen width */
        float half_sw;
        /* backend pitch */
        unsigned int pitch;
        /* backend framebuffer */
        std::uint32_t *__restrict mmio;
        /* fov scale */
        float fov_scale;
        /* 1 / fov_scale */
        float inv_fov_scale;
        /* cosine and sine of camera angle */
        float cos_cam_angle, sin_cam_angle;

        /* euclidian distance correction factor */
        std::vector<float> const* euclidian_dist_factor;
    };
}
```

rendering/sprite.h

```
#pragma once

#include "math/vec2.h"
#include "assets/ids.h"
#include "util/componentized.h"

namespace rendering {
    class software_renderer;
    class vissprite;

    class sprite : public util::componentized<sprite> {
    protected:

        /* sprite position */
        [[=util::ref_component_field{}]] math::vec2 pos;
        /* sprite height */
        [[=util::ref_component_field{}]] float z_pos;
        /* sprite angle */
        [[=util::ref_component_field{}]] float angle;
        /* sprite texture */
        assets::texture_id tex_id;
        /* texture scaling factor */
        float inherent_scale;
        /* flash time */
        float hit_flash = 0.0f;

    public:

        /* constructor */
        sprite(math::vec2 const p, float const z,
            assets::texture_id const tex, float const is);
        //sprite();
        /* virtual destructor */
        virtual ~sprite() = default;

        friend util::componentized<sprite>;
        friend software_renderer;
        friend vissprite;
    };
}
```

rendering/rendering-backend.h

```
#pragma once

#include <vector>
#include <memory>
#include <cstddef>
#include <cstdint>

#include "util/componentized.h"
#include "rendering-mode.h"

namespace rendering {
    class rendering_backend {
    public:

        /* virtual aggregator */
        virtual bool bad() const = 0;
        /* componentized-like API for mode acquisition */
        virtual std::vector<std::unique_ptr<rendering_mode const>>
            operator()(util::component_tag<"modes">) = 0;
        /* mode setting */
        virtual void push_mode(std::unique_ptr<rendering_mode const> mode) = 0;
        /* componentized-like API */
        virtual unsigned int operator()(util::component_tag<"width">) = 0;
        /* componentized-like API */
        virtual unsigned int operator()(util::component_tag<"height">) = 0;
        /* componentized-like API */
        virtual unsigned int operator()(util::component_tag<"pitch">) = 0;
        /* componentized-like API */
        virtual std::uint32_t *operator()(util::component_tag<"mmio">) = 0;
        /* wait for hardware vsync, if current mode supports it */
        virtual void wait_for_vsync() = 0;
        /* flush to display */
        virtual void flush() = 0;

        /* virtual destructor */
        virtual ~rendering_backend() = default;
    };
}
```

rendering/rendering-mode.h

```
#pragma once

#include "util/componentized.h"

namespace rendering {
    class rendering_mode : public util::componentized<rendering_mode> {
    public:
        /* virtual destructor */
        virtual ~rendering_mode() = default;

        /* componentized-like API */
        virtual unsigned int operator()(util::component_tag<"x_res">) const = 0;
        /* componentized-like API */
        virtual unsigned int operator()(util::component_tag<"y_res">) const = 0;
        /* componentized-like API */
        virtual unsigned int operator()(util::component_tag<"refresh_hz">) const = 0;
        /* componentized-like API */
        virtual bool operator()(util::component_tag<"has_vsync">) const = 0;

        friend util::componentized<rendering_mode>;
    };
}
```

rendering/software-renderer.h

```
#pragma once

#include "rendering-backend.h"
#include "frd.h"
#include "visplane.h"
#include "vissprite.h"
#include "assets/asset-manager.h"
#include "geometry/linedef.h"
#include "geometry/bsp-node.h"
#include "geometry/map-data.h"
#include "math/vec2.h"
#include <vector>
#include <cstddef>

namespace rendering {
    class software_renderer {
        /* chosen backend */
        rendering_backend &target;
        /* texture mgr reference */
        assets::asset_manager const& tex_manager;
        /* map reference */
        geometry::map_data const& current_map;

        /* near clipping plane distance */
        static constexpr float near_z = 0.1f;

        /* upper clip array */
        std::vector<int> upper_clip;
        /* lower clip array */
        std::vector<int> lower_clip;
        /* rendered visplanes */
        std::vector<visplane> visplanes;
        /* rendered vissprites */
        std::vector<vissprite> vissprites;

        /* euclidian distance correction */
        std::vector<float> euclidian_dist_factor;

        /* recursively render a BSP node */
        void render_bsp_node(util::indexed_storage<geometry::bsp_node>::id_t
            node_id, frame_rendering_data const& frd);
        /* render a linedef */
        void project_and_draw_linedef(geometry::linedef line,
            frame_rendering_data const& frd);
        /* render a wall linedef */
        void draw_solid_wall_span(float proj_x1, float proj_x2, float z1,
            float z2, float u1, float u2, geometry::linedef const& line,
            frame_rendering_data const& frd);
        /* render a portal linedef */
        void draw_portal_wall_span(float proj_x1, float proj_x2, float z1,
            float z2, float u1, float u2, geometry::linedef const& line,
            frame_rendering_data const& frd);
        /* render all visplanes */
        void render_visplanes(frame_rendering_data const& frd);
        /* queue a vissprite to be rendered */
```

```
        void add_vissprite(sprite *const s, std::uint8_t light,
                frame_rendering_data const& frd);
        /* render all vissprites */
        void render_vissprites(frame_rendering_data const& frd);
        /* whether or not a BSP should be culled based on its bounding box */
        bool is_box_visible(geometry::bsp_node::bounding_box const& box,
                frame_rendering_data const& frd);

    public:
        /* constructor */
        software_renderer(rendering_backend &tgt, assets::asset_manager const& tm, geometry::map_dat

        /* render whole bsp tree */
        void render_bsp(math::vec2 const cam_pos, float const cam_height, float cam_angle, float fov
    };
}
```

rendering/vissprite.h

```
#pragma once

#include <cstdint>
#include <vector>

#include "assets/asset-manager.h"
#include "sprite.h"

namespace rendering {
    struct frame_rendering_data;

    class vissprite {
        /* z */
        float depth;
        /* clamped screen position */
        int cx1, cx2;
        /* projected position, sprite scale */
        float proj_x, scale;
        /* projected height */
        float z_pos;
        /* sprite texture id */
        assets::texture_id tex_id;
        /* sector light level */
        std::uint8_t light_level;
        /* damage flash */
        bool flash_red;

        /* upper clip array at vissprite queue time */
        std::vector<int> upper_clip;
        /* lower clip array at vissprite queue time */
        std::vector<int> lower_clip;

    public:

        /* constructor */
        vissprite(sprite const& sprite, float const z, int const clamped_x1,
            int const clamped_x2, float const px, float const sc,
            std::uint8_t const light, std::vector<int> const& uc,
            std::vector<int> const& lc);

        /* sort for painters algorithm */
        static void pa_sort(std::vector<vissprite> &vec);

        /* render onto backend */
        void render(assets::asset_manager const& tex_manager,
            frame_rendering_data const& frd) const;
    };
}
```

rendering/visplane.h

```
#pragma once


#include <cstdint>
#include <vector>

#include "assets/asset-manager.h"
#include "assets/ids.h"

namespace rendering {
    struct frame_rendering_data;

    class visplane {
        /* visplane height */
        float height;
        /* visplane texture */
        assets::texture_id tex_id;
        /* sector light */
        std::uint8_t light_level;
        /* visplane x span */
        int min_x, max_x;
        /* visplane bounding functions */
        std::vector<int> top, bottom;

    public:

        /* constructor */
        visplane(unsigned int const sw, float const h,
            assets::texture_id const tid, std::uint8_t const light);

        /* add column to a visplane vector, either by extending an existing visplane */
        /* or by creating a new one */
        static void add_column(std::vector<visplane>& pool, int x,
            int y_start, int y_end, unsigned int sw, float flat_height,
            assets::texture_id tex_id, std::uint8_t light_level);

        /* render a visplane */
        void render(assets::asset_manager const& tex_manager,
            frame_rendering_data const& frd) const;
    };
}
```

math/ray2.h

```h
#pragma once

#include "vec2.h"
#include "segment.h"

namespace math {
    class ray2 {
    public:
        /* ray end */
        vec2 end;
        /* ray direction */
        vec2 direction;

        /* constructor */
        constexpr ray2(vec2 const e, vec2 const d);

        /* comparison operators */
        __attribute__((always_inline)) constexpr inline bool
            operator==(ray2 const other) const;
        /* comparison operators */
        __attribute__((always_inline)) constexpr inline bool
            operator!=(ray2 const other) const;

        /* rotate ray by angle */
        static ray2 const rotate(ray2 const ray, float const angle);

        /* ray intersects a segment */
        bool intersects(segment const seg, vec2 &hit_point, float &distance,
            float &seg_len) const;
    };
}

/* constexpr inline definitions */
constexpr math::ray2::ray2(vec2 const e, vec2 const d) : end(e), direction(d) {}

__attribute__((always_inline)) constexpr inline bool
math::ray2::operator==(ray2 const other) const {
    return end == other.end && direction == other.direction;
}

__attribute__((always_inline)) constexpr inline bool
math::ray2::operator!=(ray2 const other) const {
    return !(*this == other);
}
```

math/segment.h

```h
#pragma once

#include "vec2.h"
#include "util/componentized.h"

namespace math {
    class segment : public util::componentized<segment> {
        /* segment points */
        [[=util::component_field{}]] vec2 point0, point1;

    public:

        /* constructor */
        constexpr segment(vec2 const p0, vec2 const p1);

        /* comparison operators */
        __attribute__((always_inline)) constexpr inline bool
            operator==(segment const other) const;
        /* comparison operators */
        __attribute__((always_inline)) constexpr inline bool
            operator!=(segment const other) const;

        /* segment angle */
        float angle() const;
        /* segment midpoint */
        constexpr vec2 const midpoint() const;
        /* segment length */
        float len() const;
        /* segment square length */
        float sqr_len() const;

        friend util::componentized<segment>;
    };
}

/* constexpr inline definitions */
constexpr math::segment::segment(vec2 const p0, vec2 const p1) : point0(p0), point1(p1) {}

constexpr math::vec2 const math::segment::midpoint() const {
    return (point0 + point1) / 2.0f;
}

__attribute__((always_inline)) constexpr inline bool
math::segment::operator==(segment const other) const {
    return point0 == other.point0 && point1 == other.point1;
}

__attribute__((always_inline)) constexpr inline bool
math::segment::operator!=(segment const other) const {
    return !(*this == other);
}
```

math/vec3.h

```
#pragma once

#include "util/componentized.h"
#include "vec2.h"

namespace math {
    class vec3 : public util::componentized<vec3> {
        /* vector components */
        [[=util::ref_component_field{}]] float x;
        /* vector components */
        [[=util::ref_component_field{}]] float y;
        /* vector components */
        [[=util::ref_component_field{}]] float z;
    public:

        /* constructor from 3 floats */
        constexpr inline vec3(float const X = 0, float const Y = 0,
            float const Z = 0);
        /* constructor from a vec2 and height */
        constexpr inline vec3(vec2 const flat, float const Z = 0);

        /* algebraic operators */
        __attribute__((always_inline)) constexpr inline vec3 const
            operator+(vec3 const other) const;
        /* algebraic operators */
        __attribute__((always_inline)) constexpr inline vec3 const
            operator-(vec3 const other) const;
        /* algebraic operators */
        __attribute__((always_inline)) constexpr inline vec3 const
            operator-() const;
        /* algebraic operators */
        __attribute__((always_inline)) constexpr inline vec3 const
            operator*(float const scalar) const;
        /* algebraic operators */
        __attribute__((always_inline)) constexpr inline vec3 const
            operator/(float const scalar) const;

        /* algebraic operators */
        __attribute__((always_inline)) constexpr inline vec3
            &operator+=(vec3 const other);
        /* algebraic operators */
        __attribute__((always_inline)) constexpr inline vec3
            &operator-=(vec3 const other);
        /* algebraic operators */
        __attribute__((always_inline)) constexpr inline vec3
            &operator*=(float const scalar);
        /* algebraic operators */
        __attribute__((always_inline)) constexpr inline vec3
            &operator/=(float const scalar);

        /* comparison operators */
        __attribute__((always_inline)) constexpr inline bool
            operator==(vec3 const other) const;
        /* comparison operators */
        __attribute__((always_inline)) constexpr inline bool
```

```
            operator!=(vec3 const other) const;

        /* dot product of 2 vec3s */
        static constexpr float dot_product(vec3 const first, vec3 const second);
        /* cross product of 2 vec3s */
        static constexpr vec3 const cross_product(vec3 const first, vec3 const second);

        /* vector length */
        constexpr float len() const;
        /* vector square length */
        constexpr float sqr_len() const;
        /* normalized vector */
        constexpr vec3 const normalized() const;

        friend util::componentized<vec3>;
    };
}

/* constexpr inline definitions */
constexpr inline
math::vec3::vec3(float const X, float const Y, float const Z) : x(X), y(Y), z(Z) {}

constexpr inline
math::vec3::vec3(vec2 const flat, float const Z) : x(flat("x"_f)), y(flat("y"_f)), z(Z) {}

__attribute__((always_inline)) constexpr inline math::vec3 const
math::vec3::operator+(vec3 const other) const {
    return vec3(x + other.x, y + other.y, z + other.z);
}

__attribute__((always_inline)) constexpr inline math::vec3 const
math::vec3::operator-(vec3 const other) const {
    return vec3(x - other.x, y - other.y, z - other.z);
}

__attribute__((always_inline)) constexpr inline math::vec3 const
math::vec3::operator-() const {
    return vec3(-x, -y, -z);
}

__attribute__((always_inline)) constexpr inline math::vec3 const
math::vec3::operator*(float const scalar) const {
    return vec3(x * scalar, y * scalar, z * scalar);
}

__attribute__((always_inline)) constexpr inline math::vec3 const
math::vec3::operator/(float const scalar) const {
    return vec3(x / scalar, y / scalar, z / scalar);
}

__attribute__((always_inline)) constexpr inline math::vec3 &
math::vec3::operator+=(vec3 const other) {
    x += other.x; y += other.y; z += other.z;
    return *this;
}

__attribute__((always_inline)) constexpr inline math::vec3 &
math::vec3::operator-=(vec3 const other) {
```

```
    x -= other.x; y -= other.y; z -= other.z;
    return *this;
}

__attribute__((always_inline)) constexpr inline math::vec3 &
math::vec3::operator*=(float const scalar) {
    x *= scalar; y *= scalar; z *= scalar;
    return *this;
}

__attribute__((always_inline)) constexpr inline math::vec3 &
math::vec3::operator/=(float const scalar) {
    x /= scalar; y /= scalar; z /= scalar;
    return *this;
}

__attribute__((always_inline)) constexpr inline bool
math::vec3::operator==(vec3 const other) const {
    return x == other.x && y == other.y && z == other.z;
}

__attribute__((always_inline)) constexpr inline bool
math::vec3::operator!=(vec3 const other) const {
    return !(*this == other);
}

constexpr float math::vec3::dot_product(vec3 const first, vec3 const second) {
    return first.x * second.x + first.y * second.y + first.z * second.z;
}

constexpr math::vec3 const math::vec3::cross_product(vec3 const first, vec3 const second) {
    return vec3(
        first.y * second.z - first.z * second.y,
        first.z * second.x - first.x * second.z,
        first.x * second.y - first.y * second.x
    );
}
```

math/vec2.h

```
#pragma once
#include "util/componentized.h"

namespace math {
    class vec3;

    class vec2 : public util::componentized<vec2> {
        /* vector components */
        [[=util::ref_component_field{}]] float x;
        /* vector components */
        [[=util::ref_component_field{}]] float y;

    public:

        /* constructor from 2 floats */
        constexpr inline vec2(float const X = 0, float const Y = 0);
        /* constructor from a vec3, discarding z */
        vec2(vec3 const& vec);

        /* algebraic operators */
        __attribute__((always_inline)) constexpr inline vec2
            operator+(vec2 const other) const;
        /* algebraic operators */
        __attribute__((always_inline)) constexpr inline vec2
            operator-(vec2 const other) const;
        /* algebraic operators */
        __attribute__((always_inline)) constexpr inline vec2
            operator-() const;
        /* algebraic operators */
        __attribute__((always_inline)) constexpr inline vec2
            operator*(float const scalar) const;
        /* algebraic operators */
        __attribute__((always_inline)) constexpr inline vec2
            operator/(float const scalar) const;

        /* algebraic operators */
        __attribute__((always_inline)) constexpr inline vec2
            &operator+=(vec2 const other);
        /* algebraic operators */
        __attribute__((always_inline)) constexpr inline vec2
            &operator-=(vec2 const other);
        /* algebraic operators */
        __attribute__((always_inline)) constexpr inline vec2
            &operator*=(float const scalar);
        /* algebraic operators */
        __attribute__((always_inline)) constexpr inline vec2
            &operator/=(float const scalar);

        /* comparison operators */
        __attribute__((always_inline)) constexpr inline bool
            operator==(vec2 const other) const;
        /* comparison operators */
        __attribute__((always_inline)) constexpr inline bool
            operator!=(vec2 const other) const;
```

```
        /* dot product of 2 vec2s */
        static float dot_product(vec2 const first, vec2 const second);
        /* cross product of 2 vec2s */
        static float cross_product(vec2 const first, vec2 const second);

        /* vector length */
        float len() const;
        /* vector square length */
        float sqr_len() const;
        /* normalized vector */
        vec2 const normalized() const;
        /* a vector perpendicular to a given one */
        vec2 const perpendicular() const;
        /* vector argument */
        float angle() const;

        /* angle between 2 vectors */
        static float angle_between(vec2 const first, vec2 const second);
        /* vector rotation */
        static vec2 const rotate(vec2 const vec, float const angle);
        /* optimized vector rotation */
        static vec2 const rotate_with_known_trig(vec2 const vec,
            float const cos, float const sin);

        friend util::componentized<vec2>;
    };
}

/* constexpr inline definitions */
constexpr inline
math::vec2::vec2(float const X, float const Y) : x(X), y(Y) {}

__attribute__((always_inline)) constexpr inline math::vec2
math::vec2::operator+(vec2 const other) const {
    return vec2(x + other.x, y + other.y);
}

__attribute__((always_inline)) constexpr inline math::vec2
math::vec2::operator-(vec2 const other) const {
    return vec2(x - other.x, y - other.y);
}

__attribute__((always_inline)) constexpr inline math::vec2
math::vec2::operator-() const {
    return vec2(-x, -y);
}

__attribute__((always_inline)) constexpr inline math::vec2
math::vec2::operator*(float const scalar) const {
    return vec2(x * scalar, y * scalar);
}

__attribute__((always_inline)) constexpr inline math::vec2
math::vec2::operator/(float const scalar) const {
    return vec2(x / scalar, y / scalar);
}

__attribute__((always_inline)) constexpr inline math::vec2&
```

```
math::vec2::operator+=(vec2 const other) {
    x += other.x;
    y += other.y;
    return *this;
}

__attribute__((always_inline)) constexpr inline math::vec2&
math::vec2::operator-=(vec2 const other) {
    x -= other.x;
    y -= other.y;
    return *this;
}

__attribute__((always_inline)) constexpr inline math::vec2&
math::vec2::operator*=(float const scalar) {
    x *= scalar;
    y *= scalar;
    return *this;
}

__attribute__((always_inline)) constexpr inline math::vec2&
math::vec2::operator/=(float const scalar) {
    x /= scalar;
    y /= scalar;
    return *this;
}

__attribute__((always_inline)) constexpr inline bool
math::vec2::operator==(vec2 const other) const {
    return x == other.x && y == other.y;
}

__attribute__((always_inline)) constexpr inline bool
math::vec2::operator!=(vec2 const other) const {
    return !(*this == other);
}
```

util/file-resource.h

```
#pragma once
#define FILE_RESOURCE_H

#include <vector>
#include "resource.h"

namespace util {
    class file_resource : public resource {
        /* mapped file */
        std::vector<std::byte> file_data;

    public:
        /* constructor */
        explicit file_resource(std::vector<std::byte> data);
        /* destructor override with default */
        ~file_resource() override = default;

        /* componentized-like API */
        void const* operator()(component_tag<"beginning">) const override;
        /* componentized-like API */
        void const* operator()(component_tag<"end">) const override;
        /* componentized-like API */
        std::size_t operator()(component_tag<"size">) const override;
    };
}
```

util/resource.h

```
#pragma once
#define RESOURCE_H

#include <cstddef>
#include "util/componentized.h"

namespace util {
    class resource {
    public:
        /* virtual destructor */
        virtual ~resource() = default;

        /* componentized-like API */
        virtual void const* operator()(component_tag<"beginning">) const = 0;
        /* componentized-like API */
        virtual void const* operator()(component_tag<"end">) const = 0;
        /* componentized-like API */
        virtual std::size_t operator()(component_tag<"size">) const = 0;
    };
}
```

util/binary-resource.h

```
#pragma once
#define BINARY_RESOURCE_H

#include "resource.h"

namespace util {
    class binary_resource : public resource {
        /* ptr to the beginning of baked binary date */
        void const* begin_ptr;
        /* ptr to the end of baked binary date */
        void const* end_ptr;
        /* size of baked binary date */
        std::size_t byte_size;

    public:

        /* constructor */
        binary_resource(void const* begin, void const* end);
        /* destructor */
        ~binary_resource() override = default;

        /* componentized-like API */
        void const* operator()(component_tag<"beginning">) const override;
        /* componentized-like API */
        void const* operator()(component_tag<"end">) const override;
        /* componentized-like API */
        std::size_t operator()(component_tag<"size">) const override;
    };
}
```

util/componentized.h

```cpp
#pragma once
#define COMPONENTIZED_H

#include <concepts>
#include <meta>
#include <iostream>
#include <string_view>
#include <vector>

namespace util {
    /* c++26 meta annotation */
    struct component_field {};
    /* c++26 meta annotation */
    struct ref_component_field {};

    /* char array for any string size */
    template<std::size_t N>
    struct fixed_string {
        char data[N];
        /* constructor */
        consteval fixed_string(char const (&s)[N]) {
            for (std::size_t i = 0; i < N; ++i) data[i] = s[i];
        }
        /* convert to std::string_view explicitly */
        consteval std::string_view view() const { return {data, N - 1}; }
    };

    /* template metaprogramming argument struct */
    template<fixed_string field>
    struct component_tag {};

    template <typename T>
    class componentized {
        /* get a std::meta::info of a field with annotation, or {} if not found */
        template <fixed_string field, typename A>
        consteval static std::meta::info get_tagged_member() {
            auto ctx = std::meta::access_context::current();
            auto members = std::meta::nonstatic_data_members_of(^^T, ctx);

            for (auto m : members) {
                if (std::meta::identifier_of(m) == field.view()) {
                    auto attrs = std::meta::annotations_of(m);
                    for (auto attr : attrs) {
                        if (std::meta::type_of(attr) == ^^A) {
                            return m;
                        }
                    }
                }
            }
            return {};
        }

    public:
        /* auto& operator() for ref_component_field only */
        template<fixed_string field>
```

```
        requires (get_tagged_member<field, ref_component_field const>() != std::meta::info{})
        auto& operator()(component_tag<field>) {
            constexpr auto target = get_tagged_member<field, ref_component_field const>();
            return static_cast<T *>(this)->[:target:];
        }

        /* auto const& operator() for ref_component_field or component_field */
        template<fixed_string field>
        requires
            ( (get_tagged_member<field, component_field const>()     != std::meta::info{}) ||
              (get_tagged_member<field, ref_component_field const>() != std::meta::info{}) )
        auto const& operator()(component_tag<field>) const {
            constexpr auto target = []{
                auto tgt = get_tagged_member<field, component_field const>();
                if(tgt != std::meta::info{}) return tgt;
                return get_tagged_member<field, ref_component_field const>();
            }();
            return static_cast<T const *>(this)->[:target:];
        }
    };

    /* usage:
     * class test : public util::componentized<test> {
     *   [[=util::ref_component_field{}]] int hp;
     *   [[=util::component_field{}]] int max_hp;
     *
     *   float secret;
     *
     *   [[=util::component_field{}]] float speed;
     *
     *   friend class componentized<test>;
     * };
     *
     * int main() {
     *   test d(100, 5.5f);
     *
     *   std::cout << "Initial HP: " << d("hp"_f) << std::endl;
     *   d("hp"_f) = 150;
     *   std::cout << "Updated HP: " << d("hp"_f) << std::endl;
     *   std::cout << "Speed: " << d("speed"_f) << std::endl;
     * }
     */
}

/* operator""_f for the "hp"_f notation */
/* must be global */
template<util::fixed_string field>
consteval util::component_tag<field> operator""_f() {
    return {};
}
```

util/resource-loader.h

```
#pragma once

/* auto-generated file with all declarations */
#include "res.h"

#include <string>
#include <string_view>
#include <unordered_map>
#include <mutex>
#include <shared_mutex>
#include <memory>

#if __has_include(<meta>)
#include <meta>
#endif

#include "resource.h"
#include "binary-resource.h"
#include "file-resource.h"

namespace util {
    class resource_loader {
        /* resource name to pointer cache */
        std::unordered_map<std::string, void*> symbol_cache;
        /* resource name to resource cache */
        std::unordered_map<std::string, std::unique_ptr<resource>> resource_cache;
        /* cache mutex for writing */
        mutable std::shared_mutex cache_mutex;

        /* try to load from cache */
        void* lookup_binary(std::string_view resource_name);
        /* load a resource file */
        std::unique_ptr<resource> load_file(std::string_view resource_name) const;
        /* lookup baked resource via dlsym */
        resource* lookup_resource_runtime(std::string_view resource_name);

        /* turn resource name into symbol name */
        constexpr std::string make_symbol_name(std::string_view resource_name) const;

#if __has_include(<meta>)
        /* count members of the res namespace */
        static consteval std::size_t count_res_members();

        /* templated array to hold res namespace members */
        template <std::size_t N>
        struct info_array {
            std::meta::info data[N > 0 ? N : 1];
        };

        /* lookup all members of the res namespace */
        template <std::size_t N>
        static consteval info_array<N> get_res_members_array();

        /* create resource from reflection */
        constexpr resource const* get_by_symbol_reflection(std::string_view symbol_name) const;
```

```
#endif

    public:
        /* constructor */
        resource_loader()  = default;
        /* destructor */
        ~resource_loader() = default;

        resource_loader(resource_loader const&)         = delete;
        resource_loader& operator=(resource_loader const&) = delete;

        /* get resource by name */
        constexpr resource* lookup_resource(std::string_view resource_name);
    };

    /* constexpr function definitions */
    constexpr std::string resource_loader::make_symbol_name(std::string_view resource_name) const {
        std::string sym = "_binary_res_";
        sym.reserve(sym.size() + resource_name.size());

        for (char ch : resource_name) {
            unsigned char c = static_cast<unsigned char>(ch);
            if ((c >= 'a' && c <= 'z') || (c >= 'A' && c <= 'Z') || (c >= '0' && c <= '9')) {
                sym += ch;
            } else {
                sym += '_';
            }
        }
        return sym;
    }

#if __has_include(<meta>)
    consteval std::size_t resource_loader::count_res_members() {
        std::size_t count = 0;
        for (auto m : std::meta::members_of(^^res_symbols, std::meta::access_context::current())) {
            if (std::meta::has_identifier(m)) {
                if (std::string_view(std::meta::identifier_of(m)).starts_with("_binary_res_")) {
                    ++count;
                }
            }
        }
        return count;
    }

    template <std::size_t N>
    consteval resource_loader::info_array<N> resource_loader::get_res_members_array() {
        resource_loader::info_array<N> arr{};
        std::size_t idx = 0;
        for (auto m : std::meta::members_of(^^res_symbols, std::meta::access_context::current())) {
            if (std::meta::has_identifier(m)) {
                if (std::string_view(std::meta::identifier_of(m)).starts_with("_binary_res_")) {
                    if (idx < N) {
                        arr.data[idx++] = m;
                    }
                }
            }
        }
        return arr;
```

```
    }

    constexpr resource const* resource_loader::get_by_symbol_reflection(std::string_view symbol_name
        if consteval {
#if __has_include(<meta>)
            static constexpr std::size_t num_members = count_res_members();
            static constexpr auto members_arr = get_res_members_array<num_members>();

            if (!symbol_name.ends_with("_start")) {
                return nullptr;
            }
            std::string end_name(symbol_name.substr(0, symbol_name.size() - 6));
            end_name += "_end";

            unsigned char const* begin_sym = nullptr;
            unsigned char const* end_sym   = nullptr;

            /* TODO: work around a gcc 16.1.1 20260430 bug */
#pragma GCC diagnostic push
#pragma GCC diagnostic ignored "-Wshadow"
            template for (constexpr auto m : members_arr.data) {
                if constexpr (std::meta::has_identifier(m)) {
                    constexpr std::string_view name = std::meta::identifier_of(m);
                    if (name == symbol_name) {
                        begin_sym = [: m :];
                    } else if (name == end_name) {
                        end_sym = [: m :];
                    }
                }
            }
#pragma GCC diagnostic pop

            if (begin_sym && end_sym) {
                return std::define_static_object(
                    binary_resource(
                        static_cast<void const*>(begin_sym),
                        static_cast<void const*>(end_sym)));
            }
#endif
            return nullptr;
        } else {
            /* not consteval, abort */
            return nullptr;
        }
    }

    constexpr resource* resource_loader::lookup_resource(std::string_view resource_name) {
        if consteval {
            std::string begin_name = make_symbol_name(std::string(resource_name) + "_start");

            return const_cast<resource*>(get_by_symbol_reflection(begin_name));
        } else {
            return lookup_resource_runtime(resource_name);
        }
    }
#endif

}
```

util/indexed-storage.h

```
#pragma once

#include <vector>
#include <cstddef>
#include <cstdint>
#include <cassert>
#include <utility>
#include <iterator>

namespace util {
    template<typename T>
    class indexed_storage {
    public:
        /* element identifier type */
        typedef std::uint32_t id_t;

        /* sentinel identifier */
        static constexpr id_t nullid = 0;

        /* iterator entry */
        struct entry {
            id_t id;
            T& value;
        };
        /* const iterator entry */
        struct const_entry {
            id_t id;
            T const& value;
        };

        class iterator {
        private:
            /* bound indexed_storage */
            indexed_storage* storage;
            /* current index */
            size_t index;
            /* constructor */
            iterator(indexed_storage* st, size_t idx) : storage(st), index(idx) {}
            friend class indexed_storage<T>;
        public:
            /* c++ semantics */
            using iterator_category = std::forward_iterator_tag;
            /* c++ semantics */
            using value_type        = entry;
            /* c++ semantics */
            using difference_type   = std::ptrdiff_t;
            /* c++ semantics */
            using pointer           = entry*;
            /* c++ semantics */
            using reference      = entry;

            /* constructor */
            iterator() : storage(nullptr), index(0) {}

            /* access operator */
```

```
        entry operator*() const {
            return {storage->lookup[index], storage->objects[index]};
        }

        /* iteration */
        iterator& operator++() { ++index; return *this; }
        /* iteration */
        iterator operator++(int) { iterator tmp = *this; ++(*this); return tmp; }
        /* iteration */
        iterator& operator--() { --index; return *this; }
        /* iteration */
        iterator operator--(int) { iterator tmp = *this; --(*this); return tmp; }

        /* comparison operators */
        bool operator==(iterator const& other) const { return index == other.index; }
        /* comparison operators */
        bool operator!=(iterator const& other) const { return index != other.index; }
    };

    class const_iterator {
    private:
        /* bound indexed_storage */
        indexed_storage const* storage;
        /* current index */
        size_t index;
        /* constructor */
        const_iterator(indexed_storage const* st, size_t idx) : storage(st), index(idx) {}
        friend class indexed_storage<T>;
    public:
        /* c++ semantics */
        using iterator_category = std::forward_iterator_tag;
        /* c++ semantics */
        using value_type        = const_entry;
        /* c++ semantics */
        using difference_type   = std::ptrdiff_t;
        /* c++ semantics */
        using pointer           = const_entry*;
        /* c++ semantics */
        using reference      = const_entry;

        /* constructor */
        const_iterator() : storage(nullptr), index(0) {}

        /* access operator */
        const_entry operator*() const {
            return {storage->lookup[index], storage->objects[index]};
        }

        /* iteration */
        const_iterator& operator++() { ++index; return *this; }
        /* iteration */
        const_iterator operator++(int) { const_iterator tmp = *this; ++(*this); return tmp; }
        /* iteration */
        const_iterator& operator--() { --index; return *this; }
        /* iteration */
        const_iterator operator--(int) { const_iterator tmp = *this; --(*this); return tmp; }

        /* comparison operators */
```

```
        bool operator==(const_iterator const& other) const { return index == other.index; }
        /* comparison operators */
        bool operator!=(const_iterator const& other) const { return index != other.index; }
    };

protected:
    /* vector of held objects */
    std::vector<T> objects;
    /* index to id lookup */
    std::vector<id_t> lookup;
    /* id to index lookup */
    std::vector<size_t> reverse;

    /* next available id */
    id_t next_id = 1;

private:
    /* init from predefined objects */
    void init_from_objects() {
        size_t n = objects.size();
        lookup.reserve(n);
        reverse.reserve(n + 1);

        reverse.push_back(0);

        for (size_t i = 0; i < n; ++i) {
            id_t new_id = next_id++;
            lookup.push_back(new_id);
            reverse.push_back(i);
        }
    }

public:
    /* constructor */
    indexed_storage() {
        reverse.push_back(0);
    }

    /* constructor from a T vector */
    explicit indexed_storage(std::vector<T> const& initial_objects)
        : objects(initial_objects) {
        init_from_objects();
    }

    /* constructor from a T vector using move semantics */
    explicit indexed_storage(std::vector<T>&& initial_objects) noexcept
        : objects(std::move(initial_objects)) {
        init_from_objects();
    }

    /* current size */
    id_t size() const { return static_cast<id_t>(objects.size()); }

    /* add element */
    id_t add(T const& object) {
        id_t new_id = next_id++;

        if (new_id >= reverse.size()) {
```

```
        reverse.resize(new_id + 1, 0);
    }

    reverse[new_id] = objects.size();
    objects.push_back(object);
    lookup.push_back(new_id);

    return new_id;
}

/* add element with move semantics */
id_t add(T&& object) {
    id_t new_id = next_id++;

    if (new_id >= reverse.size()) {
        reverse.resize(new_id + 1, 0);
    }

    reverse[new_id] = objects.size();
    objects.push_back(std::move(object));
    lookup.push_back(new_id);

    return new_id;
}

/* remove an element */
void remove(id_t id) {
    if (id == nullid || id >= reverse.size() ||
        (reverse[id] == 0 && (objects.empty() || lookup[0] != id))) {
        return;
    }

    size_t index_to_remove = reverse[id];
    size_t last_index = objects.size() - 1;

    if (index_to_remove != last_index) {
        T& last_obj = objects.back();
        id_t last_id = lookup.back();

        objects[index_to_remove] = std::move(last_obj);
        lookup[index_to_remove] = last_id;

        reverse[last_id] = index_to_remove;
    }

    objects.pop_back();
    lookup.pop_back();
    reverse[id] = 0;
}

/* access an element */
T& operator[](id_t id) {
    assert(id != nullid && id < reverse.size() && reverse[id] < objects.size());
    return objects[reverse[id]];
}

/* access an element */
T const& operator[](id_t id) const {
```

```
            assert(id != nullid && id < reverse.size() && reverse[id] < objects.size());
            return objects[reverse[id]];
        }

        /* iteration */
        iterator begin() { return iterator(this, 0); }
        /* iteration */
        iterator end() { return iterator(this, objects.size()); }

        /* iteration */
        const_iterator begin() const { return const_iterator(this, 0); }
        /* iteration */
        const_iterator end() const { return const_iterator(this, objects.size()); }

        /* iteration */
        const_iterator cbegin() const { return const_iterator(this, 0); }
        /* iteration */
        const_iterator cend() const { return const_iterator(this, objects.size()); }

        /* remove via an iterator */
        iterator erase(iterator it) {
            if (it != end()) {
                remove((*it).id);
            }
            return it;
        }

        /* remove via a const iterator */
        const_iterator erase(const_iterator it) {
            if (it != cend()) {
                const_cast<indexed_storage*>(this)->remove((*it).id);
            }
            return it;
        }

        /* contains a specific id */
        bool contains(id_t id) const {
            return id != nullid && id < reverse.size() &&
                   (reverse[id] != 0 || (!objects.empty() && lookup[0] == id));
        }
    };
}
```

game/game.h

```cpp
#pragma once

#include <vector>
#include <memory>

#include "util/resource-loader.h"
#include "assets/asset-manager.h"
#include "input/input-backend.h"
#include "rendering/rendering-backend.h"
#include "audio/audio-backend.h"
#include "geometry/map-data.h"
#include "engine/world.h"
#include "audio/audio-mixer.h"
#include "rendering/renderer-2d.h"
#include "rendering/software-renderer.h"
#include "entities/player.h"

namespace game {
    class game {
        /* used resource loader */
        util::resource_loader rl;
        /* used asset manager */
        assets::asset_manager am;
        /* used input backend */
        std::unique_ptr<input::input_backend> input;
        /* used rendering backend */
        std::unique_ptr<rendering::rendering_backend> rb;
        /* used audio backend */
        std::unique_ptr<audio::audio_backend> ab;
        /* map */
        geometry::map_data md;
        /* audio mixer */
        audio::audio_mixer mix;
        /* 2d renderer (eg. for HUD) */
        rendering::renderer_2d r2d;
        /* 3d renderer */
        rendering::software_renderer sr;
        /* world */
        engine::world w;
        /* player pointer */
        entities::player* player_ptr;

        /* player fov */
        float fov;

        /* display a star wars like text scroll */
        void text_scroll(std::vector<std::string> const& text, assets::audio_clip_id const aid);
        /* display a text scroll from a resource */
        void ts_from_resource(std::string const& res_name, assets::audio_clip_id const aid);
        /* display the opening */
        void opening();
        /* display the ending */
        void ending();
        /* display the credits */
        void credits();
```

```
        /* show the main menu */
        void show_main_menu();
        /* show the options menu */
        void show_options();
        /* game loop */
        void loop();
        /* overlay the hud */
        void draw_hud();
    public:
        /* constructor */
        game();
        /* run game */
        void run();
    };
}
```

geometry/subsector.h

```
#pragma once

#include "util/resource.h"
#include "util/indexed-storage.h"
#include "rendering/sprite.h"
#include <vector>

namespace rendering {
    class software_renderer;
}

namespace geometry {
    class linedef;

    class subsector {
        /* lines belonging to a subsector */
        std::vector<util::indexed_storage<linedef>::id_t> lines;
        /* sprites belonging to a subsector */
        std::vector<rendering::sprite*> sprites;

    public:

        /* constructor */
        subsector() = default;
        /* constructor */
        subsector(std::vector<util::indexed_storage<linedef>::id_t> l);

        /* load subsectors from a binary */
        static std::vector<subsector> load_from_bin(util::resource const& res);

        /* remove a sprite from a subsector */
        void remove_sprite(rendering::sprite* spr);
        /* add a sprite to a subsector */
        void add_sprite(rendering::sprite* spr);

        friend rendering::software_renderer;
    };
}
```

geometry/sector.h

```
#pragma once

#include "assets/asset-manager.h"
#include "util/resource.h"
#include <vector>
#include <cstdint>

namespace rendering {
    class software_renderer;
}

namespace geometry {
    class sector {
        /* sector floor */
        float floor_height;
        /* ceiling floor */
        float ceiling_height;
        /* sector floor texture */
        assets::texture_id floor_tex;
        /* ceiling floor texture */
        assets::texture_id ceiling_tex;
        /* sector lighting */
        std::uint8_t light_level;

    public:

        /* constructor */
        sector(float fh, float ch, assets::texture_id ft, assets::texture_id ct, std::uint8_t light)

        /* load sectors from a binary */
        static std::vector<sector> load_from_bin(util::resource const& res);

        friend rendering::software_renderer;
    };
}
```

geometry/monster-spawn.h

```
#pragma once

#include <vector>
#include <cstdint>
#include "math/vec2.h"
#include "util/resource.h"
#include "util/componentized.h"
#include "entities/monster-factory.h"

namespace geometry {
    class monster_spawn : public util::componentized<monster_spawn> {
        std::uint32_t type;
        [[=util::component_field{}]] math::vec2 pos;
        float z;

        friend class util::componentized<monster_spawn>;

    public:
        monster_spawn(std::uint32_t t, math::vec2 p, float height);

        static std::vector<monster_spawn> load_from_bin(util::resource const& res);

        friend std::unique_ptr<entities::monster> entities::make_monster(geometry::monster_spawn con
    };
}
```

geometry/sidedef.h

```
#pragma once

#include "assets/ids.h"
#include "util/resource.h"
#include "util/indexed-storage.h"
#include <vector>

namespace rendering {
    class software_renderer;
}

namespace geometry {
    class sector;

    class sidedef {
        /* sector the sidedef is facing */
        util::indexed_storage<sector>::id_t facing_sector;
        /* portal upper texture */
        assets::texture_id upper_tex;
        /* wall texture */
        assets::texture_id middle_tex;
        /* portal lower texture */
        assets::texture_id lower_tex;

    public:

        /* constructor */
        sidedef(util::indexed_storage<sector>::id_t sector, assets::texture_id ut, assets::texture_i

        /* load subsectors from a binary */
        static std::vector<sidedef> load_from_bin(util::resource const& res);

        friend rendering::software_renderer;
    };
}
```

geometry/bsp-node.h

```
#pragma once

#include "math/vec2.h"
#include "util/indexed-storage.h"
#include "util/resource.h"
#include <vector>

namespace rendering {
    class software_renderer;
}

namespace geometry {
    class bsp_node {
    public:
        /* BSP node bounding box */
        struct bounding_box {
            float top, bottom, left, right;
        };

    private:

        /* partitioning line definition */
        math::vec2 pl_coord, pl_dir;
        /* front/back node bounding boxes */
        bounding_box front_box, back_box;
        /* front/back node ids */
        util::indexed_storage<bsp_node>::id_t front, back;
        /* leaf (subsector) bitmask */
        static constexpr util::indexed_storage<bsp_node>::id_t leaf_flag
            = 1 << (sizeof(util::indexed_storage<bsp_node>::id_t) * 8 - 1);

    public:

        /* is the front node a subsector */
        __attribute__((always_inline)) inline constexpr bool is_front_leaf() const;
        /* is the back node a subsector */
        __attribute__((always_inline)) inline constexpr bool is_back_leaf() const;
        /* the id of the front node */
        __attribute__((always_inline)) inline constexpr
            util::indexed_storage<bsp_node>::id_t get_front_index() const;
        /* the id of the back node */
        __attribute__((always_inline)) inline constexpr
            util::indexed_storage<bsp_node>::id_t get_back_index() const;

        /* on which side of the partitioning line does a point lie */
        bool is_pt_front_side(math::vec2 const& point) const;

        /* constructor */
        bsp_node(math::vec2 const pc, math::vec2 const pd, bounding_box const fb,
            bounding_box const bb, util::indexed_storage<bsp_node>::id_t f,
            util::indexed_storage<bsp_node>::id_t b);
        /* load subsectors from a binary */
        static std::vector<bsp_node> const load_from_bin(util::resource const& res);

        /* is any id a subsector */
```

```
        __attribute__((always_inline)) static inline constexpr bool
            is_leaf(util::indexed_storage<bsp_node>::id_t const id);
        /* literal id of a node from a possibly bitmasked one */
        __attribute__((always_inline)) static inline constexpr
            util::indexed_storage<bsp_node>::id_t
            get_id(util::indexed_storage<bsp_node>::id_t const raw);

        friend rendering::software_renderer;
    };
}

__attribute__((always_inline)) inline constexpr bool geometry::bsp_node::is_front_leaf() const {
    return is_leaf(front);
}

__attribute__((always_inline)) inline constexpr bool geometry::bsp_node::is_back_leaf() const {
    return is_leaf(back);
}

__attribute__((always_inline)) inline constexpr util::indexed_storage<geometry::bsp_node>::id_t geom
    return get_id(front);
}

__attribute__((always_inline)) inline constexpr util::indexed_storage<geometry::bsp_node>::id_t geom
    return get_id(back);
}

__attribute__((always_inline)) inline constexpr bool geometry::bsp_node::is_leaf(util::indexed_stora
    return id & geometry::bsp_node::leaf_flag;
}

__attribute__((always_inline)) inline constexpr util::indexed_storage<geometry::bsp_node>::id_t geom
    return raw & ~geometry::bsp_node::leaf_flag;
}
```

geometry/linedef.h

```
#pragma once


#include "math/segment.h"
#include "util/resource.h"
#include "util/indexed-storage.h"
#include "util/componentized.h"
#include <vector>

namespace rendering {
    class software_renderer;
}

namespace geometry {
    class sidedef;

    class linedef : public util::componentized<linedef> {
        /* line segment */
        [[=util::component_field{}]] math::segment seg;

        /* front sidedef */
        util::indexed_storage<sidedef>::id_t front;
        /* back sidedef */
        util::indexed_storage<sidedef>::id_t back;

    public:

        /* constructor */
        linedef(math::vec2 const p0, math::vec2 const p1,
            util::indexed_storage<sidedef>::id_t const f,
            util::indexed_storage<sidedef>::id_t const b);

        /* load subsectors from a binary */
        static std::vector<linedef> load_from_bin(util::resource const& res);

        /* is it a wall */
        __attribute__((always_inline)) constexpr inline bool is_wall() const;
        /* is it a portal */
        __attribute__((always_inline)) constexpr inline bool is_portal() const;
        /* line directrion */
        __attribute__((always_inline)) constexpr inline math::vec2 dir() const;
        /* line length */
        __attribute__((always_inline)) constexpr inline float len() const;

        friend util::componentized<linedef>;
        friend rendering::software_renderer;
    };
}

__attribute__((always_inline)) constexpr inline bool geometry::linedef::is_wall() const {
    return back == util::indexed_storage<sidedef>::nullid;
}

__attribute__((always_inline)) constexpr inline bool geometry::linedef::is_portal() const {
    return !is_wall();
```

```
}

__attribute__((always_inline)) constexpr inline math::vec2 geometry::linedef::dir() const {
    return (seg("point1"_f) - seg("point0"_f)).normalized();
}

__attribute__((always_inline)) constexpr inline float geometry::linedef::len() const {
    return seg.len();
}
```

geometry/pickup-spawn.h

```
#pragma once

#include <vector>
#include <cstdint>
#include <memory>
#include "math/vec2.h"
#include "util/resource.h"
#include "util/componentized.h"
#include "world_objects/pickup.h"
#include "assets/asset-manager.h"

namespace entities { class player; }
namespace audio { class audio_mixer; }
namespace engine { class world; }
namespace geometry { class pickup_spawn; }
namespace world_object {
    std::unique_ptr<world_object::pickup> make_pickup(geometry::pickup_spawn const& ps, entities::pl
}

namespace geometry {
    class pickup_spawn : public util::componentized<pickup_spawn> {
        std::uint32_t type;
        std::uint32_t subtype;
        [[=util::component_field{}]] math::vec2 pos;
        float z;

        friend class util::componentized<pickup_spawn>;

    public:
        pickup_spawn(std::uint32_t t, std::uint32_t st, math::vec2 p, float height);

        static std::vector<pickup_spawn>  load_from_bin(util::resource const& res);

        friend std::unique_ptr<world_object::pickup> world_object::make_pickup(pickup_spawn const& p
    };
}
```

geometry/map-data.h

```
#pragma once


#include "sector.h"
#include "sidedef.h"
#include "linedef.h"
#include "subsector.h"
#include "bsp-node.h"
#include "monster-spawn.h"
#include "pickup-spawn.h"
#include "rendering/sprite.h"
#include "util/indexed-storage.h"
#include "util/componentized.h"

namespace rendering {
    class software_renderer;
}
namespace game {
    class game;
}

namespace geometry {
    class map_data : public util::componentized<map_data> {
        /* all sectors */
        util::indexed_storage<sector> sectors;
        /* all sidedefs */
        util::indexed_storage<sidedef> sidedefs;
        /* all linedefs */
        [[=util::component_field{}]] util::indexed_storage<linedef> linedefs;
        /* all subsectors */
        [[=util::ref_component_field{}]] util::indexed_storage<subsector> subsectors;
        /* BSP tree */
        util::indexed_storage<bsp_node> nodes;
        /* all monster spawns */
        std::vector<monster_spawn> monster_spawns;
        /* all pickup spawns */
        std::vector<pickup_spawn> pickup_spawns;

        /* root BSP node */
        [[=util::component_field{}]] util::indexed_storage<bsp_node>::id_t root_node_id;

    public:

        /* subsector from point */
        util::indexed_storage<subsector>::id_t get_subsector_id(math::vec2 const& pt) const;

        /* move sprite to a new position */
        void move_to(rendering::sprite* spr, math::vec2 const& new_pos);

        /* load from binaries */
        static map_data load_from_bin(util::resource const& sectors_res,
            util::resource const& sidedefs_res, util::resource const& linedefs_res,
            util::resource const& subsectors_res, util::resource const& nodes_res,
            util::resource const& monsters_res, util::resource const& pickups_res);
```

```
        friend util::componentized<map_data>;
        friend rendering::software_renderer;
        friend game::game;
    };
}
```

assets/ids.h

```
#pragma once

#include <cstdint>

namespace assets {
    typedef std::int32_t texture_id;
    typedef std::int32_t menu_id;
    typedef std::int32_t audio_clip_id;
}
```

assets/audio-clip.h

```
#pragma once
#define AUDIO_CLIP_H

#include <vector>
#include <cstdint>
#include "util/resource.h"
#include "util/componentized.h"
#include "audio/audio-backend.h"

namespace assets {
    class audio_clip : public util::componentized<audio_clip> {
        /* constructor */
        audio_clip(std::vector<std::uint8_t> const &data, audio::audio_format const &fmt);

        /* PCM data to play */
        [[=util::component_field{}]] std::vector<std::uint8_t> pcm_data;
        /* supported format */
        [[=util::component_field{}]] audio::audio_format format;
        /* length in frames */
        [[=util::component_field{}]] unsigned long total_frames;

    public:
        /* load from a binary */
        static audio_clip const load_from_bin(util::resource const &res);

        friend util::componentized<audio_clip>;
    };
}
```

assets/menu.h

```
#pragma once

#include <vector>
#include <string>
#include <string_view>
#include <functional>

#include "util/resource.h"
#include "util/resource-loader.h"
#include "texture.h"
#include "menu-element.h"
#include "input/input-backend.h"
#include "ids.h"

namespace rendering {
    class rendering_backend;
    class renderer_2d;
}

namespace assets {
    class menu {
        /* constructor */
        menu(std::vector<menu_element> els, texture_id const b);

        /* menu elements */
        std::vector<menu_element> elements;
        /* menu background */
        texture_id const bg;

    public:
        /* load from binaries */
        static menu load_from_bin(util::resource_loader &resld, util::resource const &res);

        /* conditionally replace format strings with values within elements */
        /* usage: */
        /* menu.set_formatter_for("{name}", [&](std::string_view tmpl) { */
        /*   std::string s(tmpl); */
        /*   for (std::size_t p; (p = s.find("{name}")) != std::string::npos;) */
        /*       s.replace(p, 7, player_name); */
        /*   return s; */
        /* }); */
        void selective_formatter(std::string_view key_filter,
                std::function<std::string(std::string_view)> fn);

        /* add a formatter to all elements */
        void formatter_all(std::function<std::string(std::string_view)> fn);

        /* display a menu */
        int display(rendering::renderer_2d const &r2d, rendering::rendering_backend &rb,
                input::input_backend &in, std::vector<texture> const& ui_tx) const;
    };
}
```

assets/menu-element.h

```
#pragma once

#include <vector>
#include <string>
#include <functional>

#include "util/componentized.h"
#include "util/resource.h"
#include "texture.h"
#include "ids.h"

namespace rendering {
    class renderer_2d;
}

namespace input {
    struct mouse_state;
}

namespace assets {
    class menu_element {
        /* constructor */
        menu_element(int f, float bx, float by, float ex, float ey,
            texture_id const b, std::string s, int cw, int ch, std::uint8_t r,
            std::uint8_t g, std::uint8_t b_col, float rlm);

        /* function within a menu */
        int function;
        /* dimentions relative to screen size */
        float const begin_x, begin_y, end_x, end_y;
        /* element background */
        texture_id const bg;
        /* element text */
        std::string fmt_template;
        /* set formatter */
        std::function<std::string(std::string_view)> formatter;
        /* character dimentions in pixels */
        int char_w, char_h;
        /* text color */
        std::uint8_t col_r, col_g, col_b;
        /* left margin, relative to element width */
        float relative_left_margin;

        /* run template through formatter */
        std::string resolve_text() const;
        /* pack r, g, b into color */
        std::uint32_t text_color() const;
        /* set a formatter */
        void selective_fmt(std::string_view key_filter,
            std::function<std::string(std::string_view)> fn);
        /* render an element */
        int render(rendering::renderer_2d const &r2d, std::vector<texture> const& ui_tx,
                int sw, int sh, input::mouse_state const &ms) const;

    public:
```

```
        /* load from a binary */
        static menu_element load_from_bin(util::resource const &res);

        friend class menu;
    };
}
```

assets/asset-manager.h

```
#pragma once

#include <vector>
#include <string>
#include <string_view>

#include "util/resource.h"
#include "util/resource-loader.h"
#include "asset-pack.h"
#include "input/input-backend.h"
#include "ids.h"

namespace rendering {
    class renderer_2d;
    class rendering_backend;
}

namespace assets {
    class asset_manager {
        /* resource loader reference */
        util::resource_loader &rl;
        /* all loaded asset packs */
        std::vector<asset_pack> asset_packs;
        /* current asset pack */
        unsigned int cur_set;

        /* constructor */
        asset_manager(util::resource_loader &resld, std::vector<asset_pack> packs);

    public:
        /* load all assets */
        static asset_manager load(util::resource_loader &resld);

        /* next asset pack */
        void cycle_set();

        /* lookup textures */
        texture const& wall_tx_by_id(texture_id const id) const;
        /* lookup textures */
        texture const& sprite_tx_by_id(texture_id const id) const;
        /* lookup textures */
        texture const& flat_tx_by_id(texture_id const id) const;
        /* lookup textures */
        texture const& ui_tx_by_id(texture_id const id) const;
        /* lookup a menu */
        menu &menu_by_id(menu_id const id);
        /* lookup an audio clip */
        audio_clip const& audio_clip_by_id(audio_clip_id const id) const;

        /* display a menu */
        int display_menu(menu_id const id, rendering::renderer_2d const& r2d,
                rendering::rendering_backend &rb, input::input_backend &in);
    };
}
```

assets/texture.h

```h
#pragma once

#include <vector>
#include <cstddef>
#include <cstdint>

#include "util/resource.h"
#include "util/componentized.h"

namespace assets {
    class texture : public util::componentized<texture> {
        /* constructor */
        texture(std::vector<std::uint32_t> const &t, std::uint32_t const w,
                std::uint32_t const h, bool const transparent = false);

        /* pixel array */
        [[=util::component_field{}]] std::vector<std::uint32_t> pixels;
        /* texture dimentions */
        [[=util::component_field{}]] std::uint32_t width, height;

        /* whether or not the texture has transparency */
        [[=util::component_field{}]] bool has_transparency;

    public:
        /* laod from a binary */
        static texture const load_from_bin(util::resource const &res);

        friend util::componentized<texture>;
    };
}
```

assets/asset-pack.h

```h
#pragma once

#include <vector>
#include <cstddef>
#include <cstdint>
#include <string_view>
#include <unordered_map>

#include "util/resource.h"
#include "util/resource-loader.h"
#include "texture.h"
#include "menu-element.h"
#include "menu.h"
#include "audio-clip.h"
#include "ids.h"

namespace assets {
    class asset_pack {
        /* all textures */
        std::vector<texture> wall_textures;
        /* all textures */
        std::vector<texture> sprite_textures;
        /* all textures */
        std::vector<texture> flat_textures;
        /* all textures */
        std::vector<texture> ui_textures;
        /* all menus */
        std::vector<menu> menus;
        /* all audio clips */
        std::vector<audio_clip> audio_clips;

        /* constructor */
        asset_pack(std::vector<texture> walls, std::vector<texture> sprites,
            std::vector<texture> flats, std::vector<texture> ui,
            std::vector<menu> ms, std::vector<audio_clip> clips);

        /* load texures from a meta file */
        static std::vector<texture> tx_from_meta(util::resource_loader &resld,
            std::string_view meta_path);
        /* load menus from a meta file */
        static std::vector<menu> menus_from_meta(util::resource_loader &resld,
            std::string_view meta_path);
        /* load audio clips from a meta file */
        static std::vector<audio_clip> clips_from_meta(util::resource_loader &resld,
            std::string_view meta_path);
        /* parse keys in ap delaration meta file */
        static std::unordered_map<std::string, std::string>
            parse_meta_keys(std::string const& content);

    public:
        /* load from a binary */
        static asset_pack load(util::resource_loader &resld,
            std::string_view pack_meta_path);

        /* lookup textures */
```

```
        texture const& wall_tx_by_id(texture_id const id) const;
        /* lookup textures */
        texture const& sprite_tx_by_id(texture_id const id) const;
        /* lookup textures */
        texture const& flat_tx_by_id(texture_id const id) const;
        /* lookup textures */
        texture const& ui_tx_by_id(texture_id const id) const;
        /* lookup a menu */
        menu &menu_by_id(menu_id const id);
        /* lookup an audio clip */
        audio_clip const& audio_clip_by_id(audio_clip_id const id) const;

        /* display a menu */
        int display_menu(menu_id const id, rendering::renderer_2d const& r2d,
            rendering::rendering_backend &rb, input::input_backend &in);
    };
}
```

systems/health_systems.h

```
#pragma once

#include <vector>
#include <memory>
#include <algorithm>

#include "combat/status-effect.h"
#include "util/indexed-storage.h"
#include "util/componentized.h"

namespace engine {class actor;}

namespace systems {
    class health_system : public util::componentized<health_system> {
        [[=util::component_field{}]] float current_hp;
        [[=util::component_field{}]] float max_hp;
        [[=util::component_field{}]] float armor;
        [[=util::component_field{}]] float max_armor;

        [[=util::component_field{}]] std::vector<std::unique_ptr<combat::status_effect>> active_effe

        friend class util::componentized<health_system>;

    public:
        health_system(float hp, float max, float arm, float max_arm);

        void apply_damage(float amount);
        void apply_heal(float amount);
        void apply_shield(float amount);
        void apply_true_damage(float amount);

        void add_effect(std::unique_ptr<combat::status_effect> effect, engine::actor& owner);
        void process_effects(float dt, engine::actor& owner);

        bool is_dead() const;
    };
}
```

audio/alsa/backend.h

```
#pragma once

#include <alsa/asoundlib.h>
#include "audio/audio-backend.h"

namespace audio {
    namespace alsa {
        class backend : public audio::audio_backend {
        private:
            /* error check */
            bool is_bad;

            /* also pcm */
            snd_pcm_t* pcm_handle;
            /* supported format */
            audio_format fmt;
            /* supported is currently paused */
            bool paused;

            /* convert to alsa */
            snd_pcm_format_t to_alsa_format(unsigned int bits) const;

            /* configure hardware */
            bool configure_hw_params();

        public:
            /* constructor */
            backend();
            /* destructor */
            ~backend() override;

            /* set a new format */
            bool open(audio_format const& format) override;
            /* stop */
            void close() override;
            /* send pcm data */
            long write(void const* data, unsigned long frames) override;
            /* drain the pcm buffer */
            void drain() override;
            /* pause audio */
            void pause() override;
            /* resume audio */
            void resume() override;
            /* componentized-like api */
            audio_format operator()(util::component_tag<"current_format">) const override;
            /* error checking */
            bool bad() const override;
        };
    }
}
```

audio/audio-mixer.h

```
#pragma once
#define AUDIO_MIXER_H

#include "audio-backend.h"
#include "assets/audio-clip.h"
#include <vector>

namespace audio {

    class audio_mixer {
    private:
        /* one of mixed sounds */
        struct playing_sound {
            assets::audio_clip const* clip;
            unsigned long current_frame;
        };

        /* backend ref */
        audio_backend &target;
        /* supported format */
        audio_format mix_format;
        /* all currently playing sounds */
        std::vector<playing_sound> active_sounds;

    public:
        /* constructor */
        explicit audio_mixer(audio_backend &tgt, audio_format const& fmt);

        /* queue a sound */
        void play(assets::audio_clip const& clip);

        /* stop all sounds */
        void stop_all();

        /* play audio */
        void step(unsigned long frames_to_mix);

        /* are there no sounds playing */
        bool is_silent() const;
    };
}
```

audio/audio-backend.h

```cpp
#pragma once

#include <cstddef>
#include <cstdint>
#include <string>
#include <vector>

#include "util/componentized.h"

namespace audio {

    /* supported format struct */
    struct audio_format {
        unsigned int sample_rate;
        unsigned int channels;
        unsigned int bits_per_sample;
    };

    class audio_backend {
    public:
        /* virtual destructor */
        virtual ~audio_backend() = default;

        /* set a new format */
        virtual bool open(audio_format const& fmt) = 0;
        /* stop */
        virtual void close() = 0;
        /* send pcm data */
        virtual long write(void const* data, unsigned long frames) = 0;
        /* drain the pcm buffer */
        virtual void drain() = 0;

        /* pause audio */
        virtual void pause() = 0;
        /* resume audio */
        virtual void resume() = 0;
        /* componentized-like api */
        virtual audio_format operator()(util::component_tag<"current_format">) const = 0;
        /* error checking */
        virtual bool bad() const = 0;
    };
}
```

engine/actor.h

```
#pragma once

#include "renderable-entity.h"
#include "systems/health_systems.h"
#include "util/componentized.h"

namespace combat { class charmed; class slowed; }

namespace engine {

    enum class faction {
        player,
        enemy,
        neutral
    };

    class actor : public renderable_entity, public util::componentized<actor> {
    public:
        using util::componentized<actor>::operator();
        using rendering::sprite::operator();
    protected:
        /* current actor health */
        [[=util::component_field{}]] systems::health_system health;
        /* movement */
        float     movement_speed;
        /* player / enemy / neither */
        faction   team;

        friend class util::componentized<actor>;
        friend class combat::charmed;
        friend class combat::slowed;
        friend class projectile;

    public:
        /* constructor */
        actor(math::vec2 const p, float const z, assets::texture_id const tex,
            float const is, float const hp, float const shield,
            float const move_speed, faction const this_team);

        /* take damage */
        virtual void take_damage(float const dmg);
        /* take damage passing through shield */
        virtual void take_true_damage(float const dmg);
        /* heal */
        virtual void heal(float const amount);
        /* restore shield */
        virtual void add_shield(float const amount);
        /* apply an affect */
        virtual void add_effect(std::unique_ptr<combat::status_effect> effect);

        /* whether or not the actor is dead */
        bool is_dead() const;

        /* entity tick */
        virtual void update(float dt) override;
```

```
    };
}
```

engine/entity.h

```
#pragma once

namespace engine {
    class entity {
    public:
        /* tick an entity */
        virtual void update(float const dt) = 0;
        /* virtual destructor */
        virtual ~entity() = default;
    };
}
```

engine/projectile.h

```
#pragma once

#include "renderable-entity.h"
#include "actor.h"
#include "world.h"
#include "math/vec2.h"
#include "combat/status-effect.h"
#include <memory>
#include <functional>

namespace geometry { class map_data; }
namespace combat { namespace weapons { class projectile_firing_mode; } }
namespace entities { class monster; class monster_boss; }

namespace engine {

    class projectile : public renderable_entity {
        friend class combat::weapons::projectile_firing_mode;
        friend class entities::monster;
        friend class entities::monster_boss;
    protected:
        math::vec2 direction;
        float speed;
        float damage;
        float lifetime;
        float hit_radius;
        faction team;
        std::function<std::unique_ptr<combat::status_effect>()> on_hit_effect;

        world* world_ref = nullptr;
        geometry::map_data* map_ref = nullptr;
        util::indexed_storage<std::unique_ptr<entity>>::id_t self_id = 0;

    public:
        projectile(math::vec2 pos, float z, assets::texture_id tex, float scale,
                   math::vec2 dir, float spd, float dmg, float life,
                   faction f, float radius = 8.0f);

        void update(float dt) override;
    };

}
```

engine/world.h

```
#pragma once
#include <vector>
#include <memory>

#include "util/indexed-storage.h"
#include "entity.h"

namespace engine {
    class world {
    private:
        /* all entities */
        util::indexed_storage<std::unique_ptr<entity>> entities;
        /* entities to be removed at the end of a tick */
        std::vector<util::indexed_storage<std::unique_ptr<entity>>::id_t>
            deleted_entities;

    public:

        /* constructor */
        world();

        world(const world&) = delete;
        world& operator=(const world&) = delete;

        /* tick all entities */
        void update(float const dt);

        /* get an entity */
        entity& operator[](
            util::indexed_storage<std::unique_ptr<entity>>::id_t const id);
        /* get an entity */
        entity const& operator[](
            util::indexed_storage<std::unique_ptr<entity>>::id_t const id) const;
        /* all entities */
        util::indexed_storage<std::unique_ptr<entity>> const& get_entities() const;

        /* add an entity */
        util::indexed_storage<std::unique_ptr<entity>>::id_t
            register_entity(std::unique_ptr<entity> e);

        /* delete an entity */
        void delete_entity(
            util::indexed_storage<std::unique_ptr<entity>>::id_t const id);

        /* entity ptr (can be NULL) */
        entity* entity_from_id(
            util::indexed_storage<std::unique_ptr<entity>>::id_t const id) const;
    };
}
```

engine/renderable-entity.h

```
#pragma once
#include "entity.h"
#include "rendering/sprite.h"

namespace engine {
    class renderable_entity : public entity, public rendering::sprite {
    public:
        /* constructor */
        renderable_entity(math::vec2 const p, float const z, assets::texture_id const tex, float con
            : entity(), rendering::sprite(p, z, tex, is) {}

        //renderable_entity() : rendering::sprite() {}

        /* virtual destructor */
        virtual ~renderable_entity() = default;
    };
}
```

world_objects/pickup-factory.h

```h
#pragma once

#include <memory>
#include <cstdint>
#include "math/vec2.h"
#include "world_objects/pickup.h"

namespace geometry {
    class map_data;
    class pickup_spawn;
}
namespace engine { class world; }
namespace entities { class player; }
namespace audio { class audio_mixer; }
namespace assets { class asset_manager; }

namespace world_object {
    std::unique_ptr<pickup> make_pickup(geometry::pickup_spawn const& ps, entities::player& player,
}
```

world_objects/pickup.h

```
#pragma once

#include "engine/renderable-entity.h"
#include "math/vec2.h"
#include "util/indexed-storage.h"
#include "geometry/subsector.h"

namespace entities { class player; }
namespace geometry { class map_data; }

namespace world_object {

    class pickup : public engine::renderable_entity {
        float pickup_radius;
        entities::player& player_ref;
        geometry::map_data& map_ref;
        util::indexed_storage<geometry::subsector>::id_t subsector_id;
        bool consumed = false;

    public:
        pickup(math::vec2 pos, float z, assets::texture_id tex,
               entities::player& p, geometry::map_data& md,
               util::indexed_storage<geometry::subsector>::id_t sub_id,
               float radius = 20.0f, float scale = 1.0f);

        void update(float dt) override;
        bool is_consumed() const;

        virtual void on_pickup(entities::player& p) = 0;
        virtual ~pickup() = default;
    };

}
```

world_objects/weapon-pickup.h

```
#pragma once

#include <memory>
#include "pickup.h"

namespace combat { namespace weapons { class weapon; } }

namespace world_object {

    class weapon_pickup : public pickup {
        std::unique_ptr<combat::weapons::weapon> provided_weapon;
        int slot;
    public:
        weapon_pickup(math::vec2 pos, float z, assets::texture_id tex,
                      std::unique_ptr<combat::weapons::weapon> w, int weapon_slot,
                      entities::player& p, geometry::map_data& md,
                      util::indexed_storage<geometry::subsector>::id_t sub_id,
                      float radius = 25.0f, float scale = 0.4f);
        void on_pickup(entities::player& p) override;
    };

}
```

world_objects/armor-pickup.h

```
#pragma once

#include "pickup.h"

namespace world_object {

    class armor_pickup : public pickup {
        float armor_amount;
    public:
        armor_pickup(math::vec2 pos, float z, assets::texture_id tex,
                     float amount, entities::player& p,
                     geometry::map_data& md,
                     util::indexed_storage<geometry::subsector>::id_t sub_id,
                     float radius = 20.0f);
        void on_pickup(entities::player& p) override;
    };

}
```

world_objects/health-pickup.h

```
#pragma once

#include "pickup.h"

namespace world_object {

    class health_pickup : public pickup {
        float heal_amount;
    public:
        health_pickup(math::vec2 pos, float z, assets::texture_id tex,
                      float amount, entities::player& p,
                      geometry::map_data& md,
                      util::indexed_storage<geometry::subsector>::id_t sub_id,
                      float radius = 20.0f);
        void on_pickup(entities::player& p) override;
    };

}
```

world_objects/ammo-pickup.h

```
#pragma once

#include "pickup.h"
#include "entities/player.h"

namespace world_object {

    template<typename WeaponT>
    class ammo_pickup : public pickup {
        int amount;
    public:
        ammo_pickup(math::vec2 position, float z, assets::texture_id tex,
                    int amt, entities::player& pl,
                    geometry::map_data& md,
                    util::indexed_storage<geometry::subsector>::id_t sub_id,
                    float radius = 20.0f)
            : pickup(position, z, tex, pl, md, sub_id, radius), amount(amt) {}

        void on_pickup(entities::player& p) override {
            p.resupply<WeaponT>(amount);
        }
    };

}
```

combat/burning.h

```
#pragma once

#include "status-effect.h"

namespace combat
{
    class burning : public status_effect {
    public:
        burning(float const dur, unsigned int intens);

        void affect(engine::actor& target) override;
    };
}
```

combat/charmed.h

```
#pragma once

#include "status-effect.h"

namespace combat
{
    class charmed : public status_effect {
    public:
        charmed(float const dur, unsigned int intens);

        void on_apply(engine::actor& target) override;
        void on_expire(engine::actor& target) override;
    };
}
```

combat/slowed.h

```
#pragma once

#include "status-effect.h"

namespace combat
{
    class slowed : public status_effect {
    public:
        slowed(float const dur, unsigned int intens);

        void on_apply(engine::actor& target) override;
        void on_expire(engine::actor& target) override;
    };
}
```

combat/weapons/firing-mode.h

```
#pragma once

#include "math/vec2.h"

namespace combat {
    namespace weapons {
        class firing_mode {
        public:
            virtual void spawn_bullet(math::vec2 pos, float angle, float damage) = 0;
            virtual ~firing_mode() = default;
        };
    }
}
```

combat/weapons/smg.h

```
#pragma once
#include "weapon.h"

namespace geometry { class map_data; }
namespace engine { class world; }

namespace combat {
    namespace weapons {
        class smg : public weapon {
        public:
            explicit smg(geometry::map_data const& map,
                         engine::world const& world,
                         audio::audio_mixer& mix,
                         assets::asset_manager const& am);
        };
    }
}
```

combat/weapons/projectile-firing-mode.h

```
#pragma once

#include "firing-mode.h"
#include "engine/actor.h"
#include "assets/ids.h"

namespace engine { class world; }
namespace geometry { class map_data; }

namespace combat {
    namespace weapons {

        class projectile_firing_mode : public firing_mode {
            engine::world& world_ref;
            geometry::map_data& map_ref;
            engine::faction team;
            assets::texture_id projectile_tex;
            float projectile_speed;
            float projectile_lifetime;
            float projectile_scale;

        public:
            projectile_firing_mode(engine::world& w, geometry::map_data& md,
                                   engine::faction f,
                                   assets::texture_id tex, float spd,
                                   float lifetime = 5.0f, float scale = 0.5f);

            void spawn_bullet(math::vec2 pos, float angle, float damage) override;
        };

    }
}
```

combat/weapons/katana.h

```
#pragma once
#include "weapon.h"

namespace geometry { class map_data; }
namespace engine { class world; }

namespace combat {
    namespace weapons {
        class katana : public weapon {
        public:
            katana(geometry::map_data const& map, engine::world const& world,
                   audio::audio_mixer& mix, assets::asset_manager const& am);

            bool can_fire() const override;
            void fire(math::vec2 pos, float angle) override;
            void reload() override;
        };
    }
}
```

combat/weapons/hitscan-firing-mode.h

```
#pragma once

#include "firing-mode.h"
#include "geometry/map-data.h"
#include "engine/actor.h"

namespace engine { class world; }

namespace combat {
    namespace weapons {
        class hitscan_firing_mode : public firing_mode {
            geometry::map_data const& map;
            engine::world const&      world_ref;
            float                max_range;
            float                hit_radius;

        public:
            hitscan_firing_mode(geometry::map_data const& map,
                                engine::world const& world,
                                float range  = 8192.0f,
                                float radius = 32.0f);

            void spawn_bullet(math::vec2 pos, float angle, float damage) override;
        };
    }
}
```

combat/weapons/pistol.h

```
#pragma once
#include "weapon.h"

namespace geometry { class map_data; }
namespace engine { class world; }

namespace combat {
    namespace weapons {
        class pistol : public weapon {
        public:
            explicit pistol(geometry::map_data const& map,
                            engine::world const& world,
                            audio::audio_mixer& mix,
                            assets::asset_manager const& am);
        };
    }
}
```

combat/weapons/shotgun.h

```
#pragma once
#include "weapon.h"

namespace geometry { class map_data; }
namespace engine { class world; }

namespace combat {
    namespace weapons {
        class shotgun : public weapon {
        public:
            static constexpr int   pellet_count = 8;
            static constexpr float spread       = 0.2618f;

            explicit shotgun(geometry::map_data const& map,
                             engine::world const& world,
                             audio::audio_mixer& mix,
                             assets::asset_manager const& am);
            void fire(math::vec2 pos, float angle) override;
        };
    }
}
```

combat/weapons/rifle.h

```
#pragma once
#include "weapon.h"

namespace geometry { class map_data; }
namespace engine { class world; }

namespace combat {
    namespace weapons {
        class rifle : public weapon {
        public:
            explicit rifle(geometry::map_data const& map,
                           engine::world const& world,
                           audio::audio_mixer& mix,
                           assets::asset_manager const& am);
        };
    }
}
```

combat/weapons/weapon.h

```
#pragma once

#include <memory>

#include "math/vec2.h"
#include "firing-mode.h"
#include "assets/ids.h"
#include "util/componentized.h"

namespace audio { class audio_mixer; }
namespace assets { class asset_manager; }

namespace combat
{
    namespace weapons {
        class weapon : public util::componentized<weapon> {
            friend class util::componentized<weapon>;
        protected:
            std::unique_ptr<firing_mode> ammo;
            [[=util::component_field{}]] int ammo_count;
            int max_ammo;
            [[=util::component_field{}]] int reserve_mags;
            float fire_rate;
            float last_shot_time;
            float damage;
            audio::audio_mixer& mixer;
            assets::asset_manager const& assets;
            assets::audio_clip_id fire_sound_id;
            assets::audio_clip_id reload_sound_id;
            [[=util::component_field{}]] float reload_duration;
            [[=util::component_field{}]] float reload_timer;
            [[=util::component_field{}]] bool reloading;

            void finish_reload();
        public:
            virtual bool can_fire() const;
            void tick(float dt);
            void resupply(int amount);

            virtual void fire(math::vec2 pos, float angle);
            virtual void reload();
            virtual ~weapon() = default;

        protected:
            weapon(std::unique_ptr<firing_mode> firing, int max, float rate, float dmg,
                   int reserve,
                   audio::audio_mixer& mix,
                   assets::asset_manager const& am,
                   assets::audio_clip_id fire_snd = -1,
                   assets::audio_clip_id reload_snd = -1,
                   float reload_dur = 1.5f);
        };
    }
}
```

combat/weapons/sniper-rifle.h

```
#pragma once
#include "weapon.h"

namespace geometry { class map_data; }
namespace engine { class world; }

namespace combat {
    namespace weapons {
        class sniper_rifle : public weapon {
        public:
            explicit sniper_rifle(geometry::map_data const& map,
                                  engine::world const& world,
                                  audio::audio_mixer& mix,
                                  assets::asset_manager const& am);
        };
    }
}
```

combat/weapons/plasma-gun.h

```
#pragma once
#include "weapon.h"

namespace geometry { class map_data; }
namespace engine { class world; }

namespace combat {
    namespace weapons {
        class plasma_gun : public weapon {
        public:
            explicit plasma_gun(geometry::map_data& map,
                                engine::world& world,
                                audio::audio_mixer& mix,
                                assets::asset_manager const& am);
        };
    }
}
```

combat/status-effect.h

```
#pragma once

#include "math/vec2.h"

namespace engine {class actor;}

namespace combat
{
    class status_effect {
    protected:
        float duration;
        unsigned int intensity;
        float tick_interval;
        float tick_timer;
    public:
        status_effect(float const dur, float tick_inter, unsigned int intens);

        virtual void on_apply(engine::actor&);
        virtual void on_expire(engine::actor&);
        virtual void affect(engine::actor&);
        virtual ~status_effect() = default;

        bool tick(float dt);
        bool is_expired() const;
    };
}
```

entities/monster-factory.h

```
#pragma once

#include <memory>
#include <cstdint>
#include "math/vec2.h"
#include "entities/monster.h"

namespace engine { class world; }

namespace geometry {
    class monster_spawn;
    class map_data;
}

namespace entities {
    std::unique_ptr<monster> make_monster(geometry::monster_spawn const& ms, engine::actor& target,
}
```

entities/slug-projectile.h

```
#pragma once

#include "engine/projectile.h"

namespace entities {

    class slug_projectile : public engine::projectile {
    public:
        using projectile::projectile;
    };

}
```

entities/monsters/monster-magic.h

```
#pragma once

#include "entities/monster.h"

namespace entities {

    class monster_magic : public monster {
        float charge_time;
        float charge_timer;
        bool  is_charging;
        float post_fire_cooldown;
        float sidestep_timer;
        float sidestep_sign;

    public:
        monster_magic(math::vec2 const p, float const z, engine::actor& target, geometry::map_data&
        void update(float dt) override;
    };

}
```

entities/monsters/monster-elite-swift.h

```
#pragma once

#include "entities/monster.h"

namespace entities {

    class monster_elite_swift : public monster {
        float charge_speed;
        bool  is_charging;
        float charge_timer;
        float charge_cd;
        float circle_angle;
        float circle_radius;

    public:
        monster_elite_swift(math::vec2 const p, float const z, engine::actor& target, geometry::map_
        void update(float dt) override;
    };

}
```

entities/monsters/monster-basic.h

```h
#pragma once

#include "entities/monster.h"

namespace entities {

    class monster_basic : public monster {
    public:
        monster_basic(math::vec2 const p, float const z, engine::actor& target, geometry::map_data&
        void update(float dt) override;
    };

}
```

entities/monsters/monster-trapper.h

```
#pragma once

#include "entities/monster.h"

namespace entities {

    class monster_trapper : public monster {
        int   max_traps;
        int   traps_placed;
        float trap_timer;
        float trap_interval;
        float wander_timer;
        float wander_sign;

    public:
        monster_trapper(math::vec2 const p, float const z, engine::actor& target, geometry::map_data
        void update(float dt) override;
    };

}
```

entities/monsters/monster-duzy-gruby.h

```
#pragma once

#include "entities/monster.h"

namespace entities {

    class monster_Duzy_Gruby : public monster {
    public:
        monster_Duzy_Gruby(math::vec2 const p, float const z, engine::actor& target, geometry::map_d
        void update(float dt) override;
    };

}
```

entities/monsters/monster-spawner.h

```
#pragma once

#include "entities/monster.h"

namespace entities {

    class monster_spawner : public monster {
        int   max_spawns;
        int   current_spawns;
        float spawn_interval;
        float spawn_timer;

    public:
        monster_spawner(math::vec2 const p, float const z, engine::actor& target, geometry::map_data
        void update(float dt) override;
    };

}
```

entities/monsters/monster-maly-szybki.h

```
#pragma once

#include "entities/monster.h"

namespace entities {

    class monster_Maly_Szybki : public monster {
        bool  is_dashing;
        float dash_timer;
        float dash_cooldown;
    public:
        monster_Maly_Szybki(math::vec2 const p, float const z, engine::actor& target, geometry::map_
        void update(float dt) override;
    };

}
```

entities/monsters/monster-elite-tank.h

```
#pragma once

#include "entities/monster.h"

namespace entities {

    class monster_elite_tank : public monster {
        bool  melee_mode;
        float melee_threshold;
        float heavy_timer;
        float heavy_cd;
    public:
        monster_elite_tank(math::vec2 const p, float const z, engine::actor& target, geometry::map_d
        void update(float dt) override;
    };

}
```

entities/monsters/monster-ranged.h

```
#pragma once

#include "entities/monster.h"

namespace entities {

    class monster_ranged : public monster {
        float preferred_dist;
    public:
        monster_ranged(math::vec2 const p, float const z, engine::actor& target, geometry::map_data&
        void update(float dt) override;
    };

}
```

entities/monsters/monster-all-rounder.h

```
#pragma once

#include "entities/monster.h"

namespace entities {

    class monster_all_rounder : public monster {
        bool  melee_mode;
        float melee_threshold;

    public:
        monster_all_rounder(math::vec2 const p, float const z, engine::actor& target, geometry::map_
        void update(float dt) override;
    };

}
```

entities/monsters/monster-assault.h

```
#pragma once

#include "entities/monster.h"

namespace entities {

    class monster_assault : public monster {
        int   burst_size;
        float burst_interval;
        int   burst_remaining;
        float burst_timer;
        float burst_cooldown;
        float strafe_sign;
        float strafe_timer;

    public:
        monster_assault(math::vec2 const p, float const z, engine::actor& target, geometry::map_data
        void update(float dt) override;
    };

}
```

entities/monsters/monster-boss.h

```
#pragma once

#include "entities/monster.h"
#include "engine/world.h"
#include "geometry/map-data.h"
#include <vector>

namespace entities {

    class monster_boss : public monster {
        // phase tracking
        int current_phase = 1;
        float max_hp_val;

        // ranged: 4 hitscans with slow
        float ranged_cooldown = 0.0f;
        static constexpr float ranged_cd_max   = 3.0f;
        static constexpr int   projectile_count = 4;
        static constexpr float projectile_spread = 0.15f;

        // melee: burning
        // uses base attack_cooldown / attack_cd_max

        // charge (phase 2+3)
        bool  is_charging     = false;
        float charge_timer    = 0.0f;
        float charge_cooldown = 0.0f;
        float post_charge_stun = 0.0f;
        math::vec2 charge_dir{0.0f, 0.0f};
        static constexpr float charge_speed_mult = 5.0f;
        static constexpr float charge_duration   = 1.2f;
        static constexpr float charge_cd_max     = 6.0f;
        static constexpr float stun_duration     = 2.0f;

        // channel (phase 3)
        bool  is_channeling   = false;
        float channel_timer   = 0.0f;
        float flash_timer     = 0.0f;
        bool  phase3_debuffed = false;

        // spawned minions tracking
        std::vector<util::indexed_storage<std::unique_ptr<engine::entity>>::id_t> minion_ids;

        // combat stance
        bool  wants_melee    = false;
        float stance_timer   = 10.0f;

        geometry::map_data& boss_map_ref;

        float hp_ratio() const;
        void  update_phase();
        void  phase1_ai(float dt, float dist);
        void  phase2_ai(float dt, float dist);
        void  phase3_ai(float dt, float dist);
        void  ranged_attack_slow();
```

```
        void  melee_attack_burn();
        void  start_charge();
        void  update_charge(float dt);
        void  start_channel();
        bool  minions_alive() const;

    public:
        monster_boss(math::vec2 const p, float const z, engine::actor& target, geometry::map_data& m

        bool channeling() const;
        float flash_phase() const;

        void update(float dt) override;
    };

}
```

entities/monsters/monster-sniper.h

```
#pragma once

#include "entities/monster.h"

namespace entities {

    class monster_sniper : public monster {
        float shoot_interval;
        float aim_timer;

    public:
        monster_sniper(math::vec2 const p, float const z, engine::actor& target, geometry::map_data&
        void update(float dt) override;
    };

}
```

entities/plasma-projectile.h

```
#pragma once

#include "engine/projectile.h"

namespace entities {

    class plasma_projectile : public engine::projectile {
    public:
        using projectile::projectile;
    };

}
```

entities/player.h

```
#pragma once

#include <vector>
#include <memory>

#include "engine/actor.h"
#include "math/vec2.h"
#include "combat/weapons/weapon.h"
#include "util/componentized.h"

namespace geometry { class map_data; }
namespace engine { class world; }

namespace entities {
    class player : public engine::actor, public util::componentized<player> {
    public:
        using util::componentized<player>::operator();
        using engine::actor::operator();

    protected:
        [[=util::component_field{}]] std::vector<std::unique_ptr<combat::weapons::weapon>> weapons;
        float sensitivity;
        [[=util::component_field{}]] int current_weapon_index;
        geometry::map_data* map_ref = nullptr;
        engine::world* world_ref = nullptr;
        float collision_radius = 16.0f;
        float walk_speed;
        float sprint_speed;

        friend class util::componentized<player>;

    public:

        player(math::vec2 const p, float const z, assets::texture_id const tex, float const is, floa

        void equip(int slot, std::unique_ptr<combat::weapons::weapon> wpn);

        void update(float dt) override;
        void move(math::vec2 direction);
        void rotate(float yaw);
        void shoot();
        void reload();
        void switch_weapons(int index);
        void start_sprint();
        void stop_sprint();

        template<typename WeaponT>
        bool resupply(int amount) {
            for (auto& w : weapons) {
                if (!w) continue;
                if (dynamic_cast<WeaponT*>(&*w)) {
                    w->resupply(amount);
                    return true;
                }
            }
```

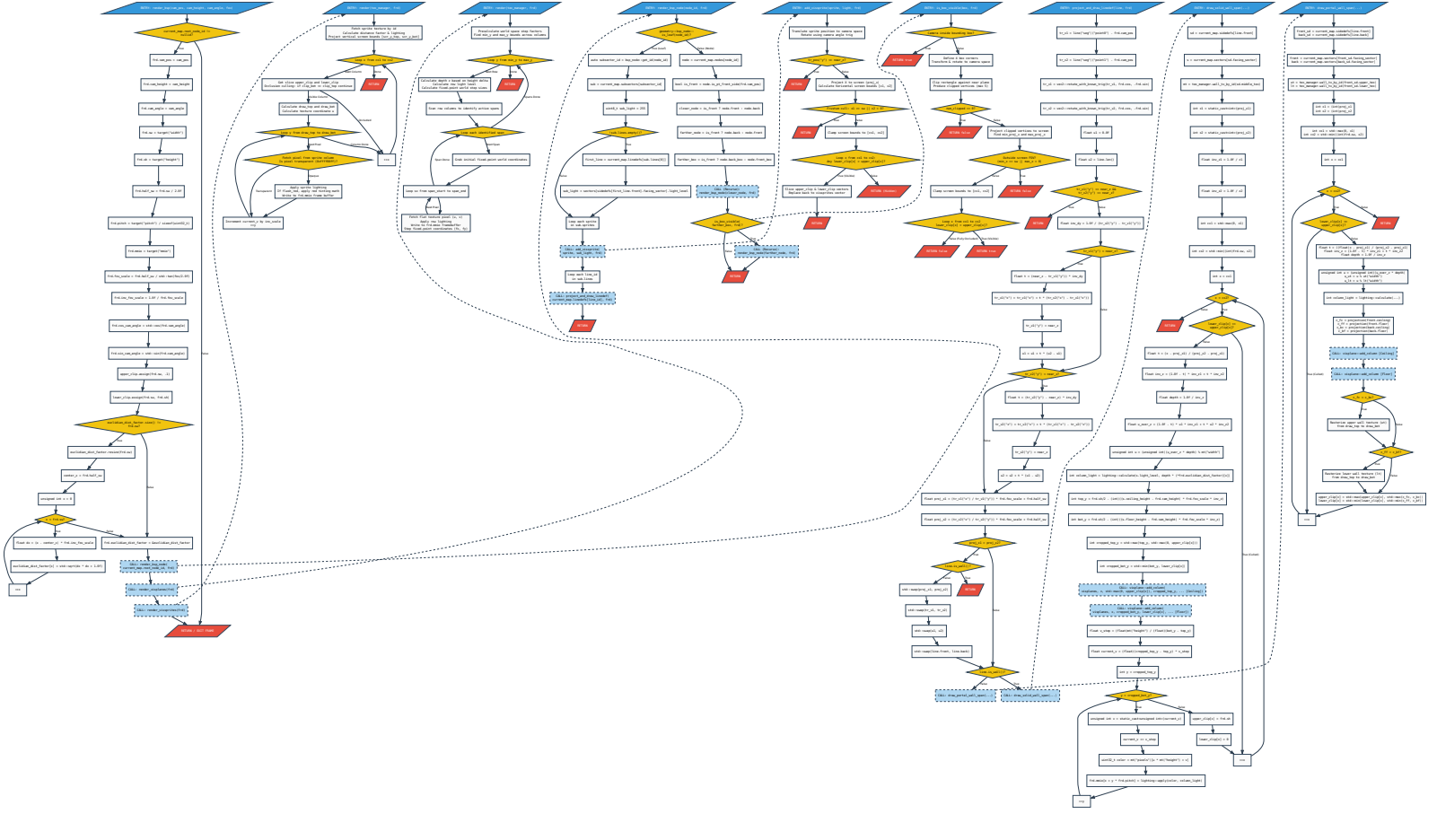
```
            return false;
        }
    };
}
```

entities/monster.h

```
#pragma once

#include "engine/actor.h"
#include "assets/ids.h"

namespace engine { class world; }
namespace geometry { class map_data; }

namespace entities {
    class monster : public engine::actor {
    protected:
        engine::actor& target;
        geometry::map_data& map_ref;
        engine::world& world_ref;
        float attack_cooldown;
        float attack_range;
        float detection_radius;
        float attack_damage;
        float attack_cd_max;
        float collision_radius = 16.0f;

        bool has_target() const;
        float dist_to_target() const;
        math::vec2 dir_to_target() const;
        void apply_wall_collision(math::vec2& new_pos) const;
        void move_toward_target(float speed, float dt);
        void move_away_from_target(float speed, float dt);
        void strafe(float speed, float dt);
        void melee_attack(float damage);
        void ranged_attack(float damage, assets::texture_id tex, float speed);

    public:

        monster(math::vec2 const p, float const z, assets::texture_id const tex, float const is, flo

        void update(float dt) override;
    };
}
```

4. Schematy blokowe własnych funkcji C++

fml


BUFFER: render_loop.ram_pos, cpu_height, ram_angle, flag
current_map.start_middle_of_it inverted
true
false
find.ram_pos = ram_pos
find.ram_height = ram_height
find.ram_angle = ram_angle
find.ac = target["width"]
find.ab = target["height"]
find.half_ac = find.ac / 2.00
find.points = target["points"] // sizeof(Int32_t)
find.data = target["data"]
find.fov_scale = find.half_ac / std::tan(fov/2.00)
find.inv_fov_scale = 1.00 / find.fov_scale
find.cos_ram_angle = std::cos(find.ram_angle)
find.sin_ram_angle = std::sin(find.ram_angle)
upper_clip.assign(find.ac, 0.0)
lower_clip.assign(find.ac, find.ab)
enviolaus_dist_factor_sized(1.0) == find.ac?
enviolaus_dist_factor(1.0)
center_x = find.half_ac
unsigned int x = 0
x < find.ac
float dx = (x - center_x) * find.inv_fov_scale
enviolaus_dist_factor(x) = std::sqrt(dx * dx + 1.00)
x++
enviolaus_dist_factor = enviolaus_dist_factor
CALL: render_playerworld()
CALL: render_cosprojection()
RETURN // NOT FOUND
BUFFER: render_tex_manager, x-ray
Finish sprite texture by id
Project vertical screen bounds (ceil_y_top, ceil_y_bot)
Loop x from x1 to x2
Get clip-clamped ceil_y_top and ceil_y_bot
Calculate wall_top and wall_bot
Loop y from ceil_y_top to wall_top
Calculate depth = based on height delta
Calculate floor_point world step sizes
Loop world floor_point coordinates
Write to framebuffer
Increment current_y by inv_scale
BUFFER: render_tex_manager, fill
Precalculate world space step factors
Loop y from sun_y to sky
Calculate depth = based on height delta
Calculate floor_point world step sizes
Scan row_colors to identify active spans
Loop each identified span
Loop world floor_point coordinates
Loop x from span_start to span_end
Fetch floor texture pixel (u, v)
Write to framebuffer
Step floor_point coordinates (du, dv)
BUFFER: render_map_node/node_ptr, fill
geometry == true_node->isPointNode_ptr?
node = current_map.nodes[node_ptr]
sub = current_map.subsectors[subsector_id]
count2 = sub_light > 255
sub_light = sectors[clamp(sidedefs[first_line, first+1].facing_sector).light_level
Loop each sprite in sub.sprites
Loop each line_id in sub.lines
CALL: render_segment(line_id)
RETURN
node = current_map.nodes[node_id]
front = is_front ? node.ls_ptr.front_side0/1?0:node.ls_ptr.front_side1?1:0 : node.back
closer_node = is_front ? node.front : node.back
farther_node = is_front ? node.back : node.front
farther_box = is_front ? node.back_box : node.front_box
CALL: fill(closer_node, map_nodes[closer_node.ptr])
is_box_visible(farther_box, true) ?
CALL: fill(farther_node, map_nodes[farther_node.ptr])
RETURN
BUFFER: app_cosprojection/sprite, light, flag
TranslateX sprite position to camera space
Rotate world camera-space into view space
tz == 0.0?
RETURN false
Project X to screen-space pixel
Calculate horizontal screen bounds (x1, x2)
x1 >= x2 || x2 < 0 || x1 >= ScreenWidth?
RETURN
Clamp screen bounds to [x1, x2]
Loop x from x1 to x2
Use 1/tz^2 for perspective correct depth
Slice upper_clip & lower_clip vectors
Replace back to cosprojtex vector
RETURN clipped
RETURN true
BUFFER: cs_bbx_visible/bbox, fill
camera outside bounding box ?
RETURN true
Retrieve 2 box corners
Transform 2 corners to camera space
Clip rectangle against near plane
Produce clipped vertices (max 5)
Project clipped vertices to screen
Find min_proj_x and max_proj_x
Outside screen FOV?
min_x >= ScreenWidth || max_x < 0
RETURN false
Clamp screen bounds to [min, ScreenWidth]
Loop x from min to max
Use lower_clip[x] < upper_clip[x]?
RETURN visible
RETURN false
BUFFER: project_and_draw_line/drline, fill
tr_x1 = line["vtg"]["point0"] - find.cam_pos
tr_y1 = line["vtg"]["point0"] - find.cam_pos
tr_x2 = line["vtg"]["point1"] - find.cam_pos
tr_y2 = line["vtg"]["point1"] - find.cam_pos
tr_x1 = cos(0-rotate_with_bonus_insight_x1, find.cos, -find.sin)
tr_y1 = sin(0-rotate_with_bonus_insight_x1, find.cos, -find.sin)
tr_x2 = cos(0-rotate_with_bonus_insight_x2, find.cos, -find.sin)
tr_y2 = sin(0-rotate_with_bonus_insight_x2, find.cos, -find.sin)
Float x1 = 0.00
Float x2 = line.len()
tr_x1 < 0.1 && tr_x2 < 0.1?
RETURN false
Float new_dy = 0.00 / (tr_x2-tr_x1) - tr_x1*(...)
tr_x1 < 0.1?
Float x1 = (tr_x1*(...) - tr_x1*(...)) / (...)
tr_x1 = 0.1
tr_x2 < 0.1?
Float x2 = (tr_x1*(...) - tr_x1*(...)) / (...)
tr_x2 = 0.1
int x1 = x1
int x2 = x2
x1 >= x2?
RETURN
Float t = (x - tr_x1)*(...) / (...)
Float inv_z = (1.00 - t) * inv_z1 + t * inv_z2
Float depth = 1.00 / inv_z
Float w_meter_z = (1.00 - t) * inv_z1 + t * inv_z2
unsigned int c = (unsigned int)(u_tex_z * depth) %...
int column_light = lightmap->calculate(...)
int top_y = find.ch/2 - (int)((tz_ceiling_height - find.cam_height) * find.fov_scale * inv_z)
int bot_y = find.ch/2 - (int)((tz_floor_height - find.cam_height) * find.fov_scale * inv_z)
int clamped_top_y = std::max(top_y, std::max(ch, upper_clip[x]))
int clamped_bot_y = std::min(bot_y, lower_clip[x])
CALL: draw_vertical_wall_column(x, clamped_top_y, clamped_bot_y, column_light, texture, u_tex_z, depth)
CALL: clip(upper_wall_column)
clip(lower_wall_column)
Float v_step = (float)tex["height"] / (float)(bot_y - top_y)
Float current_v = (float)(clamped_top_y - top_y) * v_step
int y = clamped_top_y
y < clamped_bot_y
unsigned int v = static_cast<unsigned int>(current_v)
current_v += v_step
uint32_t color = tex["pixels"][u + v*tex["height"]] | (...)
find.details.y = y * find.pitch/4 + lightmap->apply(color, column_light)
y++
BUFFER: draw_vertical_wall_column(...)
int c = current_map.sidedefs[line.front].facing_sector
int w = new_manager.wall_to_by_id(std::min(c,c), std::max(c,c))
int x1 = line[proj_x1]
int x2 = line[proj_x2]
int col = std::max(0, x1)
col < std::min(ScreenWidth, x2)
int y = col
y < ch?
upper_clip[x] < lower_clip[x]?
RETURN
Float t = (float)(y - proj_y1) / (proj_y2 - proj_y1)
Float inv_z = (1.0f - t) * inv_z1 + t * inv_z2
Float depth = 1.0f / inv_z
Float u_tex_z = (1.0f - t) * tex_u1 + t * tex_u2
unsigned int c = (unsigned int)(u_tex_z * depth) %...
int column_light = lightmap->calculate(...)
CALL: draw_vertical_wall_column(x, top_y, bot_y, column_light, texture, u_tex_z, depth)
CALL: clip(upper_wall_column)
clip(lower_wall_column)
Rendersize upper wall texture (x1) from draw_top to draw_bot
Rendersize lower wall texture (x1) from draw_top to draw_bot
upper_clip[x] = std::max(upper_clip[x], std::max(y_t, y_b))
lower_clip[x] = std::min(lower_clip[x], std::min(y_t, y_b))

5

5.1 Opis poruszania sie

5.2 Wymagania techniczne

Nazwa zasobu	Minimalna wartosc	Rekomendowana wartosc
OS	Linux 3.1	Linux 6.1
Procesor	Intel Core i3-330M 2.133 GHz	Intel Core i5-4300M 2.6 GHz
RAM	256 Mib	512 Gib
GPU	Intel HD Integrated Graphics 2000	Intel HD Integrated Graphics 4600
Storage required for compilation	512 Mib	512 Mib
Storage required for running	57 Mib	64 Mib

5.3 Uwagi instalacyjne

UWAGA: `$` i `#` jak w `man 1 bash`

5.3.1 Kompilacja programu

Aby skompilowac program, nalezy postepowac wedlug nastepujacych instrukcji:

1. Uzyc `$: git clone https://github.com/hhsNel/oop-project` lub w inny sposob zdobyc kod zrodlowy programu, a nastepnie wejsc w jego katalog
2. Usunac potencjalne pliki ktore zostaly z poprzedniej kompilacji za pomoca `$: make clean`
3. Zbudowac program za pomoca `$: make`
4. Jezeli chce sie upewnic, ze cala funkcjonalnosc dziala, nalezy przetestowac program za pomoca `$: make check`

UWAGA: Punkty 2. i 3. mozna polaczyc w jedno: `$: make clean all`. Podobnie, 3. i 4. mozna polaczyc w jedno: `$: make all check`. Podobnie, 2. 3. i 4. mozna polaczyc w jedno: `$: make clean all check`

5.3.2 Instalacja programu

Aby zainstalowac program, nalezy postepowac wedlug nastepujacych instrukcji:

1. przekopiowac zbudowany plik za pomoca `#: install isekai-doom /usr/local/bin -m 755`

UWAGA: Jezeli niedostepny jest `install`, mozna uzyc `#: cp isekai-doom /usr/local/bin && chown /usr/local/bin/isekai-doom root:root && chmod 755 /usr/local/bin/isekai-doom`

6

7

8